

Distinct catecholaminergic pathways projecting to hippocampal CA1 transmit contrasting signals during navigation in familiar and novel environments

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Abstract

Neuromodulatory inputs to the hippocampus play pivotal roles in modulating synaptic plasticity, shaping neuronal activity, and influencing learning and memory. Recently it has been shown that the main sources of catecholamines to the hippocampus, ventral tegmental area (VTA) and locus coeruleus (LC), may have overlapping release of neurotransmitters and effects on the hippocampus. Therefore, to dissect the impacts of both VTA and LC circuits on hippocampal function, a thorough examination of how these pathways might differentially operate during behavior and learning is necessary. We therefore utilized 2-photon microscopy to functionally image the activity of VTA and LC axons within the CA1 region of the dorsal hippocampus in head-fixed male mice navigating linear paths within virtual reality (VR) environments. We found that within familiar environments some VTA axons and the vast majority of LC axons showed a correlation with the animals' running speed. However, as mice approached previously learned rewarded locations, a large majority of VTA axons exhibited a gradual ramping-up of activity, peaking at the reward location. In contrast, LC axons displayed a pre-movement signal predictive of the animal's transition from immobility to movement. Interestingly, a marked divergence emerged following a switch from the familiar to novel VR environments. Many LC axons showed large increases in activity that remained elevated for over a minute, while the previously observed VTA axon ramping-to-reward dynamics disappeared during the same period. In conclusion, these findings highlight distinct roles of VTA and LC catecholaminergic inputs in the dorsal CA1 hippocampal region. These inputs encode unique information, with reward information in VTA inputs and novelty and kinematic information in LC inputs, likely contributing to differential modulation of hippocampal activity during behavior and learning.

eLife assessment

This manuscript provides **important** results that assessed the contribution of two catecholaminergic projections to the hippocampus during environment-guided reward behavior. The authors use 2-photon imaging in the hippocampus of behaving mice to provide **solid** evidence that there are dissociable roles of dopamine and norepinephrine in this structure. Although of great interest to the field of learning and memory, the results would be strengthened by additional data collected from dopaminergic projections to the hippocampus.

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Introduction

Catecholamines have a well established role in hippocampal function. Both dopamine and norepinephrine have been shown to impact hippocampal dependent learning and memory (Silva et al., 2012 [↗](#); Retailleau and Morris, 2018 [↗](#); Tsetsenis et al., 2019 [↗](#); Hansen and Manahan-Vaughan, 2014 [↗](#); Gibbs and Summers, 2002 [↗](#); Thomas, 2015 [↗](#); André et al., 2015 [↗](#)), alter synaptic plasticity (Hagena and Manahan-Vaughan, 2013 [↗](#); Hansen and Manahan-Vaughan, 2014 [↗](#); Chu et al., 2011 [↗](#); Edison and Harley, 2012 [↗](#); Hagena and Manahan-Vaughan, 2012 [↗](#); Goh and Manahan-Vaughan, 2013 [↗](#)), modulate cell excitability (Edelmann and Lessmann, 2018 [↗](#); Segal et al., 1991 [↗](#)), and influence the formation and stability of place cells (Kentros et al., 2004 [↗](#); Retailleau and Morris, 2018 [↗](#)), hippocampal neurons that selectively fire at specific locations in an environment (O'Keefe and Dostrovsky, 1971 [↗](#)). Traditionally, the main source of dopamine to the dorsal hippocampus was thought to be sparse inputs from the ventral tegmental area (VTA), while locus coeruleus (LC) inputs provided the main source of norepinephrine.

VTA DA inputs to dorsal CA1 of the hippocampus mainly innervate stratum oriens (Takeuchi et al., 2016 [↗](#); Adeniyi et al., 2020 [↗](#); Adeyelu and Ogundele, 2023 [↗](#)), and their activity bidirectionally modulates Schaffer Collateral (CA3-CA1) synapses (Rosen et al., 2015 [↗](#)), enhances persistence of reward-location associations (McNamara et al., 2014 [↗](#)), and drives place preference (Mamad et al., 2017 [↗](#)). VTA-hippocampus input activity can also bias place field location (Mamad et al., 2017 [↗](#)), improve place field stability across days (McNamara et al., 2014 [↗](#)), and drive reward expectation dependent enhancement of place field quality (Krishnan et al., 2022 [↗](#)). However, many of the effects of DA modulation of the hippocampus have now been attributed to LC inputs as their activity enhances the strength of Schaffer Collateral synapses (Takeuchi et al., 2016 [↗](#)), improves memory retention (Kempadoo et al., 2016 [↗](#)), improves place field stability across days (Wagatsuma et al., 2018 [↗](#)), and can bias place fields to a location when paired with a reward (Kaufman et al., 2020 [↗](#)) through DA mechanisms. Although many of the effects of LC and VTA are overlapping, potentially indicating shared mechanisms of action, they are believed to play different roles in spatial learning and memory (Duszkiewicz et al., 2019 [↗](#)). LC inputs influence the encoding of novel environments (Kempadoo et al., 2016 [↗](#); Wagatsuma et al., 2018 [↗](#)), while VTA DA inputs increase persistence of reward context associations (McNamara et al., 2014 [↗](#)), alter hippocampal neurons firing rate (Adeniyi et al., 2020 [↗](#)), and their suppression can evoke place avoidance (Mamad et al., 2017 [↗](#)). It is possible that these differences arise because of the differences in activity observed between LC and VTA DA neurons. Therefore, characterizing the encoding properties of LC and VTA inputs directly in the hippocampus during navigation and spatial learning would provide important insights into the specific roles of these distinct neuromodulatory pathways.

Additionally, recent findings indicate considerable heterogeneity in the activity of VTA (Engelhard et al., 2019) and LC (Uematsu et al., 2017; Noei et al., 2022; Chandler et al., 2014) neurons, highlighting the need for projection specific recordings. Therefore, we functionally imaged VTA DA and LC axons with single axon resolution in dCA1 of mice as they navigated familiar and novel virtual reality (VR) environments for rewards. We observed distinct encoding properties between these sets of inputs during navigation and in response to environmental novelty. During the approach to a previously rewarded location, the activity of most VTA DA axons ramped up. In contrast, most LC axons did not show this ramping activity and instead predicted the start of motion. Additionally, a majority of LC axons and some VTA axons showed activity associated with the animal's velocity. Following exposure to a novel environment, VTA axon ramping-to-reward signals greatly reduced but LC axon activity sharply increased. These findings support distinct roles for VTA and LC inputs to the hippocampus in spatial navigation of rewarded and novel environments.

Results

To record the activity of dopaminergic inputs to the dorsal hippocampus, we expressed axon-GCaMP6s or axon-GCaMP7b in LC or VTA neurons of different mice. We utilized the NET-cre mouse line (Wagatsuma et al., 2018) to restrict expression to noradrenergic LC neurons, and the DAT-Cre line (Zhuang et al., 2005) to restrict expression to dopaminergic VTA neurons (Fig. 1B). Mice were then head-fixed and trained to run a linear virtual reality (VR) track for water rewards delivered through a stationary waterspout when they reached the end of the virtual track (Fig. 1A). Following reward delivery, mice were teleported to the beginning of the track and allowed to complete another lap. On experiment day mice navigated the familiar, rewarded VR environment for 10 min while 2-photon microscopy was used to image the calcium activity of LC (87 ROIs from 22 imaging sites in 16 mice) or VTA (9 ROIs from 8 imaging sites in 8 mice) axons in the dorsal CA1 (Fig. 1C). Based on the z-axis depth of the recording planes, and the presence of increased autofluorescence in stratum pyramidal, we determined all 9 VTA axons were in *Stratum Oriens*, while for LC recordings, 18 sessions (78 axons in 11 mice) occurred in *Stratum Oriens* and 5 sessions (9 axons in 5 mice) in *Stratum Pyramidalis*. Example VTA (left, orange) and LC (right, blue) axon calcium activity aligned to the animal's behavior are shown in Fig. 1D. Axons from both brain regions showed periodic activity linked to the animals' exploration of the VR environment.

Distinct activity dynamics in VTA and LC inputs during rewarded navigation of a familiar environment

To examine axon activity further, we first looked at the mean activity across all axons as a function of normalized track position (Fig. 1E). As previously reported (Krishnan et al., 2022), VTA DA axons increase activity along the track, peaking at the reward location at the end of the track. In contrast, LC input activity remains relatively constant across all positions along the track (Fig. 1Eii). To examine if this difference could be due to the lower sample size of VTA axons compared to LC axons, the LC axons were down-sampled to match the VTA sample size ($n = 9$ ROIs in 8 mice) and the slope and intercepts of the down-sampled data was found. This was repeated 1000 times and did not generate any LC data points that overlap with VTA data demonstrating the difference in relationship between position and activity was not due to the different sample sizes (Fig. 1Eiii). We also examined the position related activity of individual VTA and LC axons and observed a positive relationship between position and activity in 88.9% of VTA ROIs (8/9 in 7 mice) but only 28.7% of LC ROIs (25/87 from 12 sessions in 12 mice) while 42.5% of LC ROIs (37/87 from 16 sessions in 9 mice) had a negative relationship between position and activity (Fig. 1Eiv). To account for the different track lengths between VTA and LC recordings, we also looked at the virtual distance from the rewarded end of the track as well as time from reward (Supp. Fig. 4. The same trends were seen for VTA and LC axons, with 8 VTA ROIs showing positive relationships with

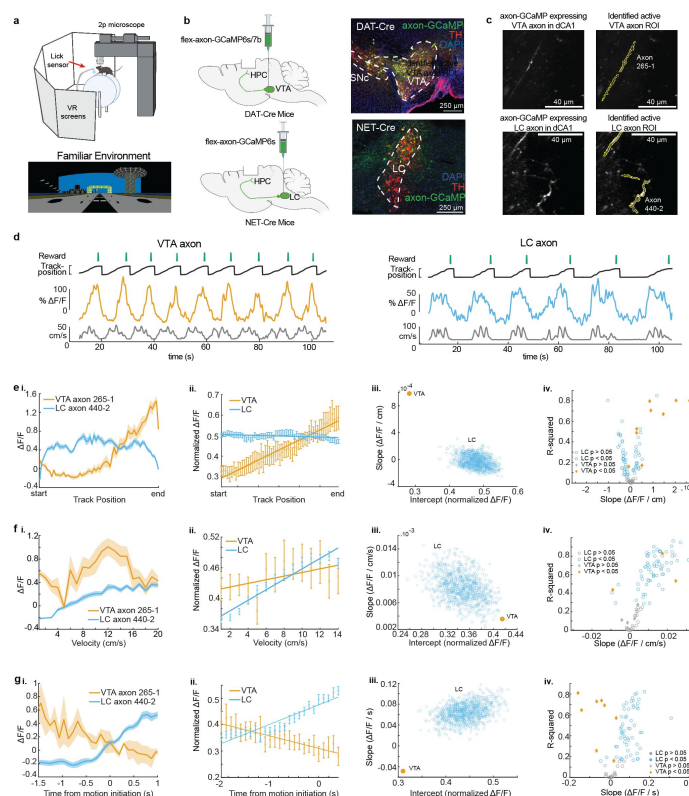


Figure 1.

Distinct activity dynamics in VTA and LC axons during navigation of familiar environments

a, Experimental setup (top), created with BioRender.com. Example virtual reality environment. **b**, Schematic representation of injection procedure (left). Representative coronal brain sections immunostained for Tyrosine Hydroxolase (TH) from a DAT-Cre mouse showing overlapping expression of axon-GCaMP (green) and TH (red) in VTA neurons (top) and from a NET-Cre mouse showing overlapping expression of axon-GCaMP (green) and TH (red) in LC neurons (bottom). **c**, Example CA1 field of view of VTA axons (top) and LC axons (bottom). Extracted regions of interest used for example VTA and LC activity throughout the figure. **d**, Example DAT-Cre mouse (left) and NET-Cre mouse (right) with aligned reward delivery (top, green), mouse track position (black), $\Delta F/F$ from example ROI (VTA-orange, LC-blue), and mouse velocity (bottom, gray). **e, i**, Position binned $\Delta F/F \pm$ s.e.m in example VTA (orange) and LC (blue) ROIs during navigation of the familiar rewarded environment. **ii**, Population average position binned $\Delta F/F \pm$ s.e.m. in VTA ROIs (orange, $n = 9$ ROIs from 8 mice) and LC ROIs (blue, $n = 87$ ROIs from 27 sessions in 17 mice). Linear regression analysis (on all data points, not means) shows that the population of VTA ROIs increase activity during approach of the end of the track while the population of LC ROIs have consistent activity throughout all positions. Linear regression, F test, VTA, $P < 1e - 21$, LC, $P < 0.01$. **iii**, The LC data set was resampled 1000x using $n = 9$ ROIs to match the number of VTA ROIs and the slope and intercept of the regression line were measured each time (blue dots). The VTA slope is steeper than all LC slopes indicating a stronger positive relationship between position and activity for VTA axons. **iv**, Linear regression of position binned activity of individual VTA (orange diamonds), and LC (blue, circles) ROIs. The majority (8/9) of VTA ROIs show a significant positive relationship with position while LC ROIs show both a positive (25/87) and negative (37/87) relationship. **f, i**, Same example ROIs as (d) binned by velocity. **ii**, Same data as (d, ii), binned by velocity. Linear regression shows that the population of VTA and LC ROIs have a significant relationship with velocity. Linear regression, F test, VTA, $P < 0.05$, LC, $P < 1e - 68$. **iii**, Resampling shows the VTA slope and intercept is within the resampled LC slopes and intercepts indicating similar relationships with velocity. **iv**, Linear regression of individual VTA and LC axons shows the majority (63/87) of LC ROIs have a significant positive relationship with velocity while only 2 VTA ROIs show this relationship. **g**, Same example ROIs as (d) aligned to motion onset. **ii**, Same data as (d, ii), aligned to motion onset. Linear regression shows that the population of VTA axons have a negative slope prior to motion onset while LC axons have positive slope. Linear regression, F test, VTA, $P < 0.01$, LC, $P < 1e - 65$. **iii**, Resampling shows the VTA slope is negative while all resampled LC slopes are positive. **iv**, Linear regression of individual VTA and LC ROIs shows the majority (56/87) of LC ROIs have a significant positive slope prior to motion onset while the majority (6/9) of VTA ROIs have a negative slope.

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both distance and time from reward, and 21 LC ROIs showing a positive relationship with distance from reward and 31 LC ROIs showing a positive relationship with time to reward. This suggests that track length does not influence the encoding properties of VTA and LC axons.

For a subset of VTA ($n = 6$) and LC ROIs ($n = 26$), the reward at the end of the track was removed, and the activity of these axons in the unrewarded condition was examined (**Fig. 2A**). While the slope of the VTA population activity across positions significantly decreased (**Fig. 2B.i**) as expected and previously reported (Krishnan et al., 2022), the LC population activity across positions did not significantly change (**Fig. 2C.i**). This confirmed that the ramping activity in VTA axons was due to the animal's proximity to an expected reward and this signal was not present in the average activity of LC axons.

Next, we investigated the mean activity of these axons as a function of velocity. The population mean of both VTA and LC axons increased as velocity increased (**Fig. 1F**). This is consistent with the finding that LC inputs to dCA1 encode velocity (Kaufman et al., 2020) and the finding that some DA VTA neurons encode kinematics (Engelhard et al., 2019). Again, to account for differences in sample size, we down sampled the LC axons 1000 times and found the slope and y-intercept of each sampling. The overlap of the VTA and LC slopes and intercepts confirms we cannot conclude any differences in velocity related activity in the VTA and LC axon populations (**Fig. 1F.iii**). However, analyzing individual ROIs, we observed a statistically significant positive relationship between velocity and activity in the majority, 72.4%, of LC axons (63/87 ROIs from 21 sessions in 13 mice), while only 28.6% (2/9 ROIs in 2 mice) of VTA axons showed a positive relationship with velocity (**Fig. 1F.iv**). The strong velocity correlated activity in a small subset of VTA DA axons indicates heterogeneity in the activity of these inputs similar to what is observed in VTA DA somas (Engelhard et al., 2019). To determine if the velocity correlated activity in VTA DA axons is confounded by the ramping to reward activity, we examined this activity in the unrewarded condition where the ramping to reward activity is absent. Here, the VTA population activity across all velocities is decreased but the slope, or the relationship between velocity and activity, remains unchanged (**Fig. 2B.ii**). This is consistent with a subset of VTA axons encoding velocity information, while the overall decrease in activity is explained by the decrease in reward related activity demonstrated in **Fig. 2B.i**. However, the relationship between velocity and LC activity is unchanged in the unrewarded environment (**Fig. 2C.ii**), indicating condition invariant velocity encoding in LC axons.

Finally, we examined the activity of LC and VTA axons during rest and the transition to movement. The population of LC axons ramped up in activity during the 2 s leading up to motion onset (**Fig. 1G**). This is consistent with reports of activity of LC axons in cortical areas (Reimer et al., 2016) showing LC activity prior to motion onset. In contrast, VTA axons show decreasing activity during the 2 s leading up to motion onset (**Fig. 1G**). This ramping down in VTA axon activity is likely due to most periods of immobility occurring between reward delivery and the start of the next lap, during which we previously demonstrated reward related activity decays in VTA axons (Krishnan et al., 2022). Indeed, in the unrewarded condition the negative slope of the VTA population activity leading up to motion onset disappears (**Fig. 2B.iii**), indicating this relationship is largely driven by the presence of reward rather than motion onset. However, the LC population activity remains unchanged in the unrewarded environment (**Fig. 2C.iii**), suggesting this activity is related to the lead up to motion initiation. These differences in activity are not an artifact of lower sample size of VTA ROIs as shown by down-sampling the LC ROIs activity 1000 times and measuring the slopes and intercepts of the down-sampled data did not generate any data points that overlapped with the VTA slope and intercept (**Fig. 1G.iii**). In further support of distinct activity profiles leading up to motion onset, we found that the majority (6/9 ROIs in 6 mice) of VTA ROIs decreased in activity leading up to motion onset but only 2/87 LC ROIs in one session decreased in activity, while the majority, (56/87 ROIs in 17 sessions in 9 mice), of LC ROIs increased in activity leading up to motion onset (**Fig. 1G.iv**). Together, this analysis demonstrates overlapping but distinct activity in VTA and LC axons during spatial navigation with

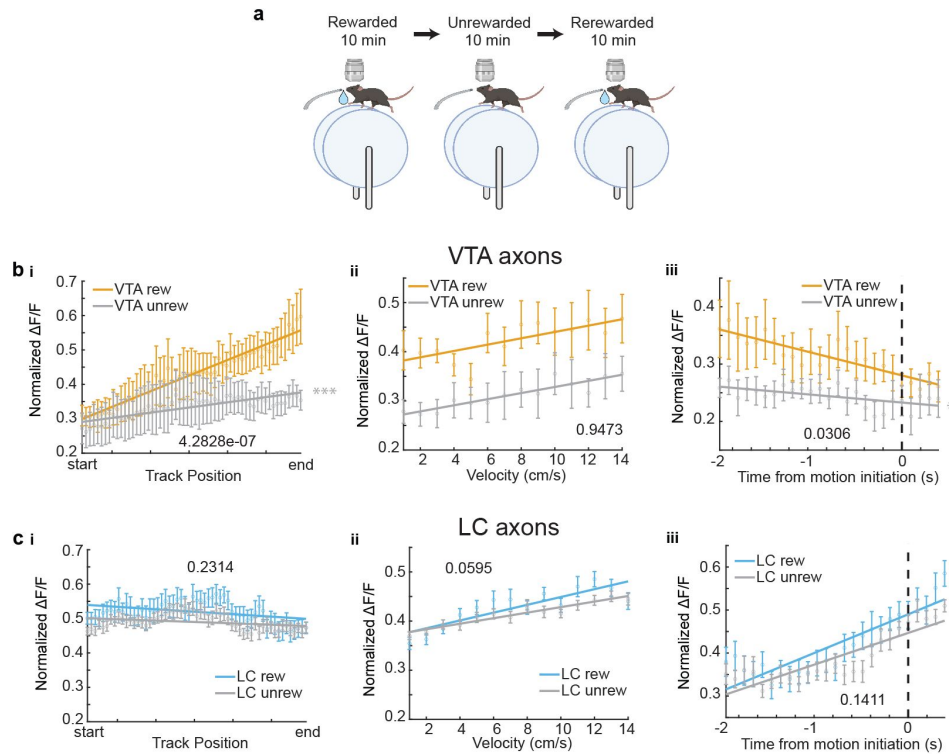


Figure 2.

Removal of reward restructures VTA but not LC input activity during spatial navigation:

a. Experimental Paradigm. **b.i.** Population average position binned $\Delta F/F \pm$ s.e.m. of VTA ROIs ($n = 6$ ROIs in 6 mice) in the rewarded (VTA rew, orange) and unrewarded conditions (VTA unrew, gray). Linear regression, F test, Rewarded, $P < 1e - 22$, Unrewarded, $P < 0.001$. **ii.** Same data as (b,i) binned by velocity. Linear regression, F test, Rewarded, $P < 0.05$, Unrewarded, $P < 0.05$. **iii.** Same data as (b,i) aligned to motion onset. Linear regression, F test, Rewarded, $P < 1e - 4$, Unrewarded, $P < 0.05$. **c, i,** Population average position binned $\Delta F/F \pm$ s.e.m. of LC ROIs ($n = 26$ ROIs in 7 sessions in 4 mice) in the rewarded (LC rew, blue) and unrewarded conditions (LC unrew, gray). Linear regression, F test, Rewarded, $P < 0.01$, Unrewarded, $P < 0.01$. **ii.** Same data as (c,i) binned by velocity. Linear regression, F test, Rewarded, $P < 1e - 12$, Unrewarded, $P < 1e - 15$. **iii.** Same data as (c,i) aligned to motion onset. Linear regression, F test, Rewarded, $P < 1e - 21$, Unrewarded, $P < 1e - 26$. The slope of each unrewarded measure was compared to the familiar laps using a one-way ANCOVA with Tukey HSD post hoc test. * $P < 0.05$, ** $P < 0.001$, *** $P < 1e - 4$.

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VTA axons showing strong activity correlated with distance to reward and some velocity correlated activity, while LC axons demonstrate activity correlated to velocity and time to motion onset.

Because the DAT-Cre mice were injected with a virus for expression of either axon-GCaMP6s (4 mice) or axon-GCaMP7b (4 mice), we asked whether these two GCaMP variants led to different activity dynamics (Supp. Fig. 2). The axon-GCaMP6s (5 ROIs) and axon-GCaMP7b (4 ROIs) both had a positive relationship with position, negative relationship with time to motion initiation, and a neutral relationship with velocity, although the GCaMP6s relationships were stronger (Supp. Fig. 2 A, C). Importantly, when we compared only axon-GCaMP6s expressing VTA ROIs with the axon-GCaMP6s expressing LC ROIs (Supp. Fig. 3), we saw similar relationships as the comparisons using all VTA ROIs (**Fig. 1** [↗](#)).

Environmental Novelty induces activity in LC but not VTA inputs

Both LC and VTA neurons have been shown to respond to novel environments (Takeuchi et al., 2016 [↗](#)). Therefore, we aimed to examine the activity of VTA DA and LC inputs to the hippocampus during exposure to novel VR environments. Following 10 minutes in the familiar environment, mice were teleported to a novel VR environment of the same track length, with a reward at the same position at the end of the track. Following teleportation, we found the running speed of both DAT-Cre and NET-Cre mice transiently decreased (**Fig. 3A** [↗](#)) and they spend less time immobile (**Fig. 3A** [↗](#)), demonstrating mice recognize they are navigating a novel environment. While the velocity quickly recovered for both groups of mice, the freezing ratio, or amount of time spent immobile, never recovered in the first ten laps in the novel environment for the NET-Cre mice indicating some novelty induced changes in behavior persist.

We aligned VTA and LC axon activity to the switch to the novel environment and investigated changes in activity due to exposure to novelty. To test whether the mean axon activity is significantly elevated or lowered, we defined a baseline by generating 1000 shuffles of the axon traces across the entire recording sessions, downsampling the shuffled data 1000 times to match the VTA ($n = 7$) and LC ($n = 87$) sample sizes, and calculating the mean and 95% CI of the shuffled data. After teleportation, the periodic activity observed in the mean of VTA axons, likely reflecting ramping-to-reward signals in each individual axon, disappeared (**Fig. 3B** [↗](#)). This is evident in the traces of most of the individual VTA ROIs showing a loss of the ramping-to-reward signal (**Fig. 3C** [↗](#)), and in the VTA population position binned activity showing a significant reduction in ramping activity (Supp. Fig. 1). However, 1 VTA ROI showed an increase in activity immediately following exposure to novelty (**Fig. 3C** [↗](#)), indicating heterogeneity across VTA axons in CA1 and the lack of a novelty signal on average may be due to a small sample size.

Strikingly, LC axons show a dramatic increase in mean activity that remained elevated for > 1 minute following exposure to the novel environment (**Fig. 3B** [↗](#)) similar with findings that LC cell body activity is elevated for minutes following exposure to environmental novelty (Takeuchi et al., 2016 [↗](#)). A significant increase in activity above baseline activity following the switch to a novel environment can be seen in 36 LC axon ROIs (from 15/22 sessions in 10/16 mice) (**Fig 3C** [↗](#)). To further characterize this activity, we found the mean population activity for each lap and separately for 10 s time bins leading up to and following exposure to the novel environment. This analysis shows that LC activity is significantly elevated above baseline for 6 consecutive laps and approximately 90 seconds following exposure to the novel environment (**Fig. 3D** [↗](#)). These findings demonstrate that LC inputs signal environmental novelty, supporting a role for these inputs in novelty encoding in the hippocampus.

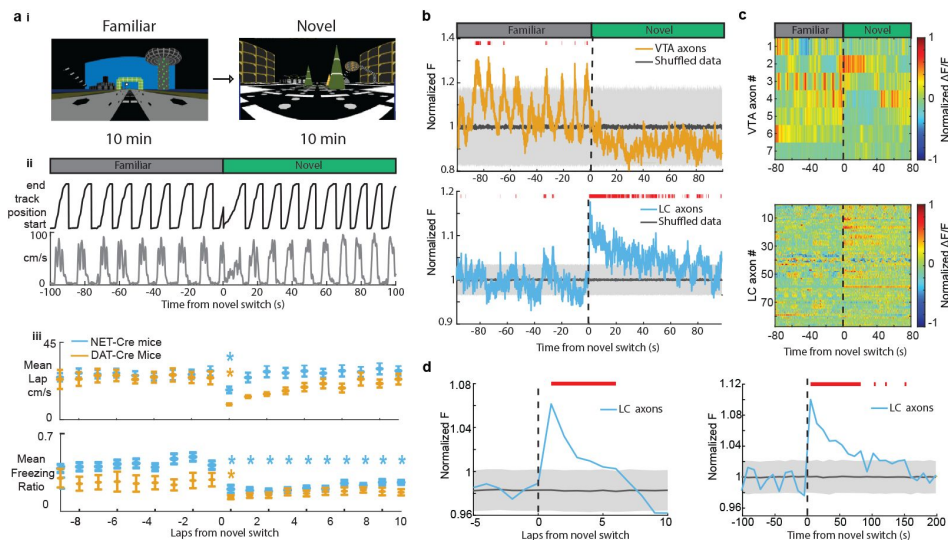


Figure 3.

Exposure to a novel environment causes an abrupt and sustained increase in activity in LC but not VTA inputs to dCA1

a, i, Experimental Paradigm. ii, Behavior from example mouse during the transition from the familiar VR environment to a novel VR environment showing the animals track position (top, black) and velocity (bottom, gray). iii, The average running velocity $\pm$ s.e.m. of all mice during the transition to a novel VR environment (top). The average freezing ratio of all mice $\pm$ s.e.m., calculated as the time spent immobile (velocity < 5 cm/s) divided by the total lap time. Each lap was compared to the final lap in the familiar environment using a one-way ANOVA with Tukey HSD post hoc test (Blue *, NET-Cre mice $P < 0.05$; Orange *, DAT-Cre mice $P < 0.05$). b, Mean normalized fluorescence of all VTA ROIs (top, $n = 7$) and LC ROIs (bottom, $n = 87$) aligned to the switch to the novel environment. To define a baseline and 95% CI (gray shaded region), 1000 shuffles were created from the calcium traces and down sampled to match the sample size and averaged. This was repeated 1000 times and the mean and 95% CI of this shuffled data was determined for each frame. Red lines indicate periods where two or more consecutive frames passed above the 95% CI of the shuffled baseline. c, Normalized $\Delta F/F$ activity of all VTA ROIs (top) and LC ROIs (bottom) aligned to the switch to the novel VR environment. d, The normalized fluorescence of all LC ROIs binned by lap (left) or into 50 frame bins (right). The baseline and 95% CI (gray shaded region) was defined using the same method as in (b). Red lines indicate bins above the baseline 95% CI.

Novelty-induced changes in behavior explains the late but not early increases in LC activity

It is possible that the change in the amount of time the animals spend running versus immobile in the novel environment could explain the increase in LC activity in the novel environment, as LC activity is related to behavior (**Fig. 1**). For instance, LC axons show elevated activity during motion versus rest (**Fig. 1G**). Therefore, an increase in the time spent in motion upon exposure to the novel environment could lead to an increase in LC activity. To account for the differences in time spent running vs immobile between the two environments, we removed any periods where the mice were immobile to isolate the effects of novelty from changes in behavior (**Fig. 4B**). When only looking at activity during running in both environments, we found that LC axon activity is elevated for 2 laps, or 40 s, in the novel environment (**Fig. 4B-E**). Additionally, when we look only at activity when the mice were immobile, we see that activity is elevated for 30s in the novel environment (Supp. Fig. 5 B). Together, this indicates that there are two separate components that drive LC axon activity during the initial exposure to the novel environment. One, a shorter purely novelty-induced increase in activity which occurs during the first 2 laps, or about 40 s, in the novel environment. Two, a behavior-induced increase in LC activity caused by an increase in the percentage of time spent running that extends beyond the increase in the novelty-induced activity for 6 laps or 90 s.

If the short, novelty-induced signal in LC axons is an additional signal riding on top of the behavior correlated signals - position, velocity, and motion onset we would expect a disruption of these behavioral correlations during the initial lap in a novel environment. To test this we examined these behavioral correlations lap-by-lap following exposure to the novel environment. Indeed, the slopes of the position binned, velocity binned, and motion onset aligned data are all significantly more negative in the first lap in the novel environment than the final laps of the familiar environment (**Fig. 4F-H**). This is consistent with a decaying novelty signal that peaks at the start of the first lap and rides on top of these behaviorally-correlated signals. This produces an elevation in activity at positions near the start of the first lap that is lower at positions near the end of the first lap, causing a negative slope relationship between position and LC axon activity on the first lap (**Fig. 4F**; light green line). Further, low velocities occur at the start of each lap compared to the end of the lap. Because the novelty signal is highest when animals are running slowest, the novelty signal flattens the velocity-LC activity relationship (**Fig. 4G**; light green line). Lastly, rest periods typically occur at the start of the track. Therefore, motion onset encoding on the first lap in the novel environment occurs when the novelty signal is highest, again, flattening the relationship (**Fig. 4H**; light green line). By the third lap in the novel environment, where the novelty-induced signal is no longer observable, the relationships between LC activity and position is no longer different than the relationship in the familiar environment (**Fig. 4F**, -green). The relationship between LC activity and motion onset is also no longer different by the third lap in the novel environment (**Fig. 4H**). Although the relationship between velocity and LC activity is different in the third novel lap than that of the final familiar laps, by the final lap in the novel environment it is no longer different than the familiar relationship (**Fig. 4G**). Together this demonstrates that the relationships between LC activity and behavior in the novel environment quickly return to those in the familiar environment. Interestingly, the slope is significantly increased in the final lap of the novel environment (**Fig. 4F**), potentially indicating the development of activity at the novel reward location as has been previously reported (Kaufman et al., 2020). Altogether, examining the lap-by-lap dynamics of the position, velocity, and motion onset activity indicates that environmental novelty induces a sharp increase in LC input activity during the first two laps in the novel environment while also inducing a change in behavior that leads to increased LC input activity in subsequent laps.

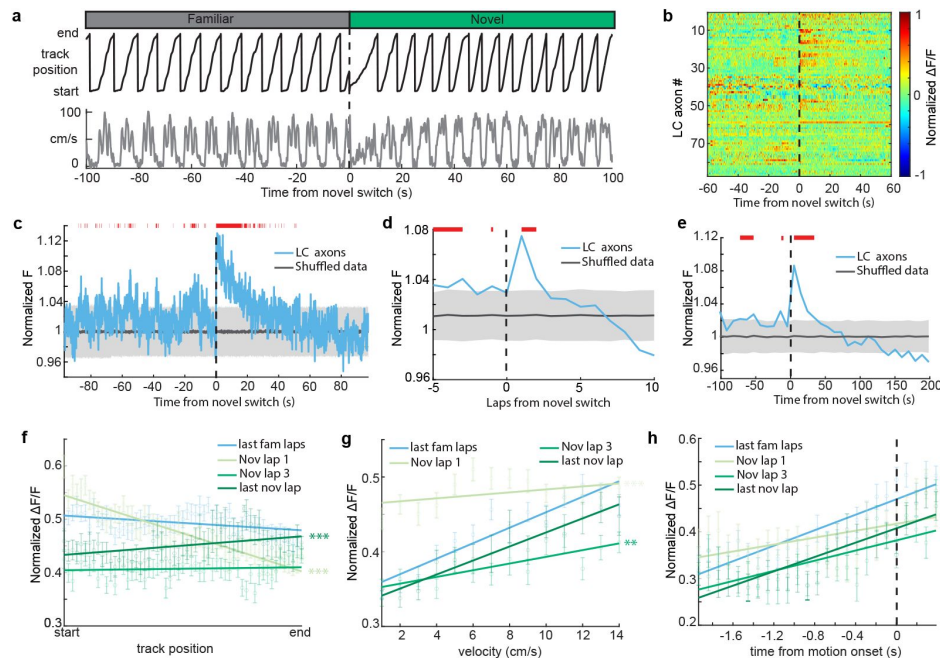


Figure 4.

Novelty-induced changes in behavior explain the late but not early increases in LC activity

a, Good behavior from example mouse following removal of freezing periods (velocity < 0.2 cm/s) during the transition from the familiar to a novel VR environment showing the animals track position (top, black) and velocity (bottom, gray). b, Normalized $\Delta F/F$ activity of all LC ROIs aligned to the switch to the novel VR environment following removal of freezing periods. c, Mean normalized F of all LC ROIs (bottom, $n = 87$) aligned to the switch to the novel environment. To define a baseline and 95% CI (gray shaded region), 1000 shuffles were created from the calcium traces and down sampled to match the sample size and averaged. This was repeated 1000 times and the mean and 95% CI of this shuffled data was determined for each frame. Red lines indicate periods where two or more consecutive frames passed above the 95% CI of the shuffled baseline. d,e., The normalized F of all LC ROIs binned by lap (d) or into 50 frame bins (e). The baseline and 95% CI (gray shaded region) was defined using the same method as in (c). Red lines indicate bins above the baseline 95% CI. f-h, Population average position binned (f), velocity binned (g), and motion onset aligned (h) $\Delta F/F \pm$ s.e.m. in LC ROIs ($n = 90$) in the final laps of the familiar environment (light blue), and the first (orange), third (dark blue), and final laps (green) of the novel environment. Linear regression, F test, position binned (f) fam last, $P < 1e-4$, nov 1, $P < 1e-68$, nov 3, $P = 0.48$, nov last, $P < 0.001$; velocity binned (g) fam last, $P < 1e-31$, nov 1, $P < 0.05$, nov 3, $P < 1e-5$, nov last, $P < 1e-13$; motion onset aligned (h) fam last, $P < 1e-29$, nov 1, $P < 0.001$, nov 3, $P < 1e-9$, nov last, $P < 1e-18$. The slope of each lap was compared to the final familiar laps using a one-way ANCOVA with Tukey HSD post hoc test. * $P < 0.01$, ** $P < 0.001$, *** $P < 1e-4$.

We also examined the activity of a subset of LC axons (50 ROIs from 11 sessions in 9 mice) during the transition from darkness to the familiar VR environment. These axons showed elevated activity for 20s following initial exposure to the familiar environment (Supp. Fig. 5 A). While this activity did not persist as long as the activity following exposure to the novel environment, it indicates these axons may be generally responsive to abrupt and unpredicted changes to the animal's environment. This fits with the proposed role of the LC in arousal, with novelty driving greater arousal than abrupt exposure to a familiar environment.

Discussion

During spatial navigation in a familiar environment, activity of VTA DA inputs to dCA1 were strongly modulated by position relative to reward, ramping up as mice approached the end of the track where reward was located. This activity could be related to the animal's distance or time from the rewarded location. Further experiments should be conducted to distinguish between time and distance such as those conducted in VTA DA soma recordings (Kim et al., 2020 [link](#)). We have previously shown that this activity is dependent on the history of reward delivery and reflects the animals' reward expectation (Krishnan et al., 2022 [link](#)). VTA axons in dCA1 also showed decreasing activity during rest prior to motion onset. However, after removing reward, VTA axons showed no activity prior to motion onset indicating this relationship was driven by reward related activity. This ramping activity cannot be explained by changes in the animals' sensory experience, as the VR environment and waterspout position remained unchanged in the unrewarded condition. Additionally, on the initial laps with no reward, the ramping activity is still present (Krishnan et al., 2022 [link](#)) indicating this activity is not directly related to the delivery of water but is instead caused by the animal's internal state of reward expectation.

On average, VTA axons showed activity modulated by velocity in a familiar rewarded environment. This relationship was largely due to the activity of two VTA axons that were strongly modulated by velocity, suggesting that there is heterogeneity in the population of VTA axons in dCA1. A positive relationship between average VTA axon activity and velocity persisted in the unrewarded condition and in the novel environment, indicating velocity encoding in these inputs that is not a result of reward related activity. This heterogeneity in the encoding across individual VTA axons is consistent with studies demonstrating heterogeneous encoding of behavioral variables in VTA DA cell bodies, including activity related to rewards and kinematics (Engelhard et al., 2019 [link](#)).

Our findings that VTA DA axons show no novelty induced activity and instead show reduced activity following exposure to a novel environment, is in contrast with several studies showing novelty induced activity in VTA DA cell bodies (Takeuchi et al., 2016 [link](#); Duszkievicz et al., 2019 [link](#); Lisman and Grace, 2005 [link](#)), indicating potential heterogeneity in VTA neurons in response to novelty. It is possible that some VTA DA inputs to dCA1 respond to novel environments, and the small number of axons recorded here are not representative of the whole population. Another possibility is that the lack of a novelty response we observe is due to differences in experimental design. Here, mice learned to approach a location for reward which has been shown to lead to ramping activity in dopaminergic VTA neurons (Howe et al., 2013 [link](#); Krishnan et al., 2022 [link](#); Kim et al., 2020 [link](#); London et al., 2018 [link](#); Jeong et al., 2022 [link](#)). Following exposure to novelty, the disappearance of reward related activity could obscure any novelty induced increases in activity. In addition, as noted above, on average we did observe a velocity associated signal in VTA axons. When mice were exposed to the novel environment their velocity initially decreased. This would be expected to reduce the average signal across the VTA axon population relative to the higher velocity in the familiar environment. It is possible that this decrease could somewhat mask a subtle novelty induced signal in VTA axons. Therefore, additional experiments should be conducted to investigate the heterogeneity of these axons and their activity under different experimental conditions during tightly controlled behavior.

LC axons showed no position encoding. Instead, they were modulated by velocity and ramped up in activity prior to motion initiation, consistent with recordings of LC axons by others in dCA1 (Kaufman et al., 2020) and in the cortex (Reimer et al., 2016), respectively. An important question is how is this LC axon activity impacting hippocampal neurons during navigation in a familiar environment? It is possible that LC axons during navigation provide increases in excitability that promotes place cell activity as both dopamine and norepinephrine in the hippocampus can impact cell excitability (Segal et al., 1991; Edelmann and Lessmann, 2011, 2018). Additionally, place cells are flexible during spatial navigation with new place fields forming in familiar environments (Sheffield et al., 2017; Dong et al., 2021) and shifting position with time/experience (Dong et al., 2021). Place fields can also shift to follow changing reward (Gauthier and Tank, 2018) and object locations (Bourboulou et al., 2019). Dopamine and norepinephrine have also been shown to impact hippocampal synaptic plasticity (Zhang et al., 2009; Edelmann and Lessmann, 2011; Hagena and Manahan-Vaughan, 2012; Goh and Manahan-Vaughan, 2013). Therefore, LC inputs may promote the plasticity necessary for place cells to flexibly adapt to changes in a familiar environment (Kaufman et al., 2020; Redondo and Morris, 2011). In other words, LC inputs could allow the hippocampus to be flexible during navigation through their impacts on synaptic plasticity (Takeuchi et al., 2016; Yamasaki and Takeuchi, 2017; Duzskiewicz et al., 2019).

Exposure to environmental novelty leads to an increase in dopamine in the dorsal hippocampus (Ihalainen et al., 1999) and promotes synaptic plasticity (Li et al., 2003; Hagena and Manahan-Vaughan, 2012), hippocampal replay (McNamara et al., 2014; Dupret et al., 2010) and memory persistence (Li et al., 2003; Cohen et al., 2017). In our experiment, exposure to a novel environment caused an increase in LC axon activity but not in VTA DA axon activity, supporting findings that novel experiences induce activity of LC neurons (Takeuchi et al., 2016). As discussed above, the slowing down of animal behavior in the novel environment could have decreased LC axon activity and reduced the magnitude of the novelty signal we detected during running. The novelty signal we report here may therefore be an under estimate of its magnitude under matched behavioral settings. The increased activity of LC neurons in the novel environment could increase hippocampal neuron activity (Wagatsuma et al., 2018), increase efficacy of Schaffer Collateral synapses (Takeuchi et al., 2016), stabilize place cells across days (Wagatsuma et al., 2018) and enhance memory persistence (Wagatsuma et al., 2018; Takeuchi et al., 2016; Chowdhury et al., 2022) through dopamine receptor dependent mechanisms. Importantly, while LC inputs to CA1 have been shown to cause an increase in activity (Wagatsuma et al., 2018; Chowdhury et al., 2022), shape over-representation of novel reward locations (Kaufman et al., 2020), and modulate memory linking (Chowdhury et al., 2022), they have not been shown to play a role in the formation of contextual memories (Chowdhury et al., 2022) or stabilization of place cell maps across days (Wagatsuma et al., 2018) in dCA1. However, it is possible this novelty induced activity in LC inputs to dCA1 impacts the formation of instant place fields observed in novel environments (Sheffield et al., 2017; Dong et al., 2021). Instant place fields form on the first lap of a novel environment, right when the LC novelty signal is highest in dCA1, suggesting LC inputs may play a role in their formation or stabilization. Therefore, further experiments should investigate the role of LC axons on dCA1 place fields on a trial-by-trial basis.

While LC neurons have been shown to impact novelty encoding through dopaminergic mechanisms (Wagatsuma et al., 2018; Takeuchi et al., 2016; Chowdhury et al., 2022), this does not exclude the possibility that they also release norepinephrine during exposure to novelty and exploration of a familiar environment. Indeed, hippocampal levels of norepinephrine also increase during exposure to environmental novelty (Lima et al., 2019; Moreno-Castilla et al., 2017), but how this norepinephrine release effects hippocampal function is not well understood. Additionally, it is not known whether norepinephrine and dopamine are released from the same LC inputs or from distinct sets of LC inputs. Dopamine is in the synthesis pathway of norepinephrine and in LC neurons it is loaded into vesicles where it is then converted to norepinephrine by dopamine β -hydroxylase (Cimarrusti et al., 1979). It is possible that high levels

of activity of LC inputs, like those occurring during exposure to novelty, lead to release of vesicles before dopamine can be converted to norepinephrine thus leading to the release of dopamine under these conditions. However, low levels of LC activation, like those observed during familiar environment exploration, may provide time for dopamine to be converted to norepinephrine and thus lead to the release of norepinephrine from LC terminals under these conditions. Further experiments investigating the dynamics of dopamine conversion and release from LC terminals in the hippocampus should be conducted to test this hypothesis.

Here we show that LC input activity is modulated by velocity, time to motion onset, abrupt exposure to a familiar environment and abrupt exposure to novel environments. Each of these conditions is associated with an increase in arousal, and LC activity has been strongly linked to arousal levels (Aston-Jones and Bloom, 1981 [↗](#); Berridge and Waterhouse, 2003 [↗](#); Carter et al., 2010 [↗](#)). Therefore, rather than encoding each of these variables independently, LC inputs are likely encoding the animals arousal level during spatial navigation. It has been shown that attention and arousal levels impact tuning properties in many cortical areas and this is thought to be mediated through LC activity (Shulman et al., 1979 [↗](#); Bouret and Sara, 2002 [↗](#); Martins and Froemke, 2015 [↗](#); Waterhouse and Navarra, 2019 [↗](#)). Similarly, changes in the animals' brain state, including changes in attention (Kentros et al., 2004 [↗](#)) and engagement (Pettit et al., 2022 [↗](#)), alter the tuning properties of place cells. This indicates arousal could impact the function of hippocampal neurons through these LC inputs, either directly or through astrocytes (Peter Rupprecht et al., 2023 [↗](#)).

The distinct activity dynamics exhibited by LC and VTA DA axons during spatial navigation of familiar and novel environments underscore their distinct contributions to hippocampal dependent learning and memory processes. Notably, these findings reinforce the notion that VTA DA inputs play a pivotal role in the ongoing maintenance and updating of associations between expected rewards and the locations that lead to them, while LC axons appear to be integral to the process of encoding memories of entirely new environments and stimuli.

Methods

Subjects

All experimental and surgical procedures were in accordance with the University of Chicago Animal Care and Use Committee guidelines. For this study, we used 10–20 week-old male Slc6a3Cre⁺/– (DAT-Cre⁺/–) mice obtained from JAX labs and Slc6a2 Cre⁺/– (NET-Cre⁺/–)(23–33 g) mice obtained from the Tonegawa lab (Wagatsuma et al., 2018 [↗](#)). Male mice were used over female mice due to the size and weight of the headplates (9.1 mm × 31.7 mm, 2 g) which were difficult to firmly attach on smaller female skulls. Mice were individually housed in a reverse 12 h light/dark cycle at 72 °F and 47% humidity, and behavioral experiments were conducted during the animal's dark cycle.

Mouse surgery and viral injections

Mice were anesthetized (1–2% isoflurane) and injected with 0.5 ml of saline (intraperitoneal injection) and 0.05 ml of Meloxicam (1–2 mg/kg, subcutaneous injection) before being weighed and mounted onto a stereotaxic surgical station (David Kopf Instruments). A small craniotomy (1–1.5 mm diameter) was made over the ventral tegmental area (VTA) (± 0.5 mm lateral, 3.1mm caudal of Bregma) of DAT-Cre⁺/– mice or over the locus coeruleus (LC) (± 0.875 mm lateral, -5.45 mm caudal of Bregma). The genetically-encoded calcium indicator, pAAV-hsyn-Flex-Axon-GCaMP6s (pAAV-hSynapsin1-FLEX-axon-GCaMP6s was a gift from Lin Tian (Addgene viral prep # 112010-AAV5 ; <http://n2t.net/addgene:112010> [↗](#) ; RRID:Addgene 112010) was injected into the VTA of DAT-Cre⁺/– mice (200 nL at a depth of 4.4 mm below the surface of the dura) or the LC of NET-Cre⁺/– mice (200 nL at a depth of 3.65 mm below Bregma). For a subset (4/8) of VTA recordings, a different GCaMP

variant, pAAV-Ef1A-Flex-Axon-GCaMP7b, was injected due to the difficulty finding and recording VTA axons in dCA1 (pAAV-Ef1A-Flex-Axon-GCaMP7b (pAAV-Ef1A-Flex-Axon-GCaMP7b was a gift from Rylan Larsen - Addgene plasmid # 135419; <http://n2t.net/addgene:135419> ; RRID: Addgene 135419). Following injections, the site was covered up using dental cement (Metabond, Parkell Corporation) and a metal head-plate (9.1 mm × 31.7 mm, Atlas Tool and Die Works) was also attached to the skull with the cement. To reduce bleeding and swelling during the hippocampal window implantation, mice were separated into individual cages and water restricted for 3 weeks (0.8–1.0 ml per day). Mice then underwent surgery to implant a hippocampal window as previously described (Dombeck et al., 2010 ). Following implantation, the headplate was attached with the addition of a head-ring cemented on top of the head-plate which was used to house the microscope objective and block out ambient light. Post-surgery mice were given 2–3 ml of water/day for 3 days to enhance recovery before returning to the reduced water schedule (0.8–1.0 ml/day).

Behavior and virtual reality

Our virtual reality (VR) and treadmill setup was designed similar to previously described setups (Sheffield et al., 2017 ; Heys et al., 2014 ). The virtual environments that the mice navigated through were created using VIRMEn (Aronov and Tank, 2014 ). 2 m (DAT-Cre mice) or 3 m (NET-Cre mice) linear tracks rich in visual cues were created that evoked numerous place fields in mice as they moved along the track at all locations (Fig. 1  (Bourboulou et al., 2019 ). Mice were head restrained with their limbs comfortably resting on a freely rotating styrofoam wheel (‘treadmill’). Movement of the wheel caused movement in VR by using a rotatory encoder to detect treadmill rotations and feed this information into our VR computer, as in refs. (Sheffield et al., 2017 ; Heys et al., 2014 ). Mice received a water reward (4 µL) through a waterspout positioned directly in front of the animal’s mouth upon completing each traversal of the track (a lap), which was associated with a clicking sound from the solenoid. Licking was monitored by a capacitive sensor attached to the waterspout. Upon receiving the water reward, a short VR pause of 1.5 s was implemented to allow for water consumption and to help distinguish laps from one another rather than them being continuous. Mice were then virtually teleported back to the beginning of the track and could begin a new traversal. Mouse behavior (running velocity, track position, reward delivery, and licking) was collected using a PicoScope Oscilloscope (PICO4824, Pico Technology, v6.13.2). Behavioral training to navigate the virtual environment began 4–7 days after window implantation (30 min per day) and continued until mice reached >2 laps per minute, which took 10–14 days (although some mice never reached this threshold). Mice that reached this behavioral threshold and had adequate GCaMP expression were imaged the following day. Out of the 36 NET-Cre mice injected, 15 were never recorded for either failing to reach behavioral criteria, or a lack of visible expression in axons. Out of the 54 DAT-Cre mice injected, images were never conducted in 36 for lack of expression or failing to reach behavioral criteria.

Two-photon imaging

Imaging was done using a laser scanning two-photon microscope (Neurolabware). Using a 8 kHz resonant scanner, images were collected at a frame rate of 30 Hz with bidirectional scanning through a 16x/0.8 NA/3 mm WD water immersion objective (MRP07220, Nikon). GCaMP6s and GCaMP7b were excited at 920 nm with a femtosecond pulsed two photon laser (Insight DS + Dual, Spectra-Physics) and emitted fluorescence was collected using a GaAsP PMT (H11706, Hamamatsu). The average power of the laser measured at the objective ranged between 50–80 mW. A single imaging field of view (FOV) between 400–700 µm equally in the x/y direction was positioned to collect data from as many VTA or LC axons as possible. Time-series images were collected from 3–5 planes spaced 2 µm apart using an electric lens to ensure axons remained in a field of view and reduce power going to an individual plane. Images were collected using Scanbox (v4.1, Neurolabware) and the PicoScope Oscilloscope (PICO4824, Pico Technology, v6.13.2) was used to synchronize frame acquisition timing with behavior.

Imaging sessions

The familiar environment was the same environment that the animals trained in. The experiment protocol for single day imaging sessions is shown in [Fig. 1](#). Each trial lasted 8–12 min and was always presented in the same order. Mice were exposed to the familiar rewarded environment for 10 minutes, then were immediately teleported to the start of a novel rewarded VR environment and allowed to navigate for 10 minutes. Mice on average ran 19 ± 3.8 (mean $\pm$ 95% CI) laps in the familiar environment, at which point they were teleported to the novel environment and imaging continued for 30 ± 5.4 laps. The Novel-rewarded environment (N) had distinct visual cues, colors and visual textures, but the same dimensions (2 or 3 m linear track) and reward location (end of the track) as the familiar environment. Imaging sessions with large amounts of drift or bleaching were excluded from analysis (8 sessions for NET mice, 6 sessions for LC Mice).

Histology and brain slices imaging

We checked the expression axon-GCaMP to confirm expression was restricted to the VTA of DAT-Cre mice and the LC of NET-Cre. Mice were deeply anesthetized with isoflurane and perfused with 10 ml phosphate-buffered saline (PBS) followed by 20 mL 4% paraformaldehyde in PBS. The brains were removed and immersed in 30% sucrose solution overnight before being sectioned at 30 μ m-thickness on a cryostat. Brain slices were collected into well plates containing PBS. Slices were washed 5 times with PBS for 5 min then were blocked in 1% Bovine Serum Albumin, 10% Normal goat serum, 0.1% Triton X-100 for 2 h. Brain slices were then incubated with 1:500 rabbit- α -TH (MAB318, Sigma Aldrich) and 1:500 mouse- α -GFP (SAB2702197, Sigma Aldrich) in blocking solution at 4 °C. After 48 h, the slices were incubated with 1:1000 goat- α -rabbit Alexa Fluor 647nm secondary antibody (A32731, ThermoFisher) and 1:1000 goat- α -mouse Alexa Fluor 488nm (A32723, ThermoFisher) for 2 h. Brain slices were then collected on glass slides and mounted with a mounting media with DAPI (SouthernBiotech DAPI-Fluoromount-G Clear Mounting Media, 010020). The whole-brain slices were imaged under $\times 10$ and $\times 40$ with a Caliber I.D. RS-G4 Large Format Laser Scanning Confocal microscope from the Integrated Light Microscopy Core at the University of Chicago.

Image processing and ROI selection

Time-series images were preprocessed using Suite2p (v0.10.1)79. Movement artifacts were removed using rigid and non-rigid transformations and assessed to ensure absence of drifts in the z-direction. Datasets with z-drifts were discarded, as determined by visually assessing imaging sessions, followed by using the registration metrics output by suite2p (8 sessions). For axon imaging, ROIs were first defined using Suite2p and manually inspected for accuracy. ROIs were then hand drawn over all segments of Suite2p defined active axons using ImageJ to ensure all axon segments were included for analysis. Fluorescent activity for each ROI was extracted and highly correlated ROIs (Pearson correlation coefficient ≥ 0.7) were combined and their fluorescent activity was extracted. To be included Baseline corrected $\Delta F / F$ traces across time were then generated for each ROI using both a small window of 300 frames for lap by lap analysis, and a larger sliding window of 2000 frames to avoid flattening slow signals for novelty response analysis. Additional ROIs were drawn over autofluorescent structures that were not identified by suite2p. These “blebs” were processed in the same way as axon ROIs and used as controls to check for imaging and motion artifacts.

To remove low signal to noise axons, we defined the SNR of each ROI using the power spectrum of their fluorescent activity similar to ([Reimer et al., 2016](#)). For frequencies above 1Hz, the power was defined as noise because this sits outside of the range of frequencies possible for GCaMP6s fluorescence. The SNR ratio was then defined as the ratio of the peak power between 0.5 Hz and

1Hz over the average power between 1 Hz and 3 Hz. The SNR of “blebs” was also determined and any axon with a SNR greater than 1.5 std from the mean of the “blebs” SNR was used for analysis (110/231 LC ROIs, 9/9 VTA ROIs).

Additionally, it was observed that a subset of axon ROIs would greatly increase fluorescence at seemingly random time points and remain elevated for the rest of the trial. This activity could be due to the axons being unhealthy and filling with calcium. Therefore, we identified these axons using the *cusum* function in matlab to detect changes in mean activity that remained elevated for at least 2000 frames or at least 500 frames if they were still elevated at the end of the recording session and removed them from analysis (23/110 LC ROIs, 0/7 VTA ROIs).

Behavioral Analysis

Mouse velocity was calculated as the change in VR position divided by the sampling rate and smoothed using a Savitzky-Golay filter with a 7 frame window and 5 degree polynomial. To find the lap mean velocity, periods where the mice were immobile (velocity < 5 cm/s) were removed and the average velocity during the remaining frames was calculated. The lap mean freezing ratio was calculated as the number of frames spent immobile (velocity < 5 cm/s) divided by the total number of frames for each lap.

Axon Imaging Analysis

For the three measures below, to avoid weighting axons with a high SNR more than others each ROI was normalized by $(\Delta F / F - \Delta F / F \text{ min}) / (\Delta F / F \text{ max} - \Delta F / F \text{ min})$ where $\Delta F / F \text{ min}$ is the 1st quantile and $\Delta F / F \text{ max}$ was the 99 quantile for each ROI. The 1st and 99th quantiles were used in order to avoid normalize to noisy outlier data points.

Position binned fluorescence

To find the position binned fluorescent activity of each ROI, the track was divided into 5 cm bins. For each lap, the average fluorescence in each bin was calculated for each ROI. The position binned fluorescence was then averaged across all laps in each environment to find the mean position binned activity in the familiar and novel environments.

Velocity binned fluorescent activity

To find the velocity binned fluorescent activity for each ROI, the velocity was divided into 1 cm/s bins from 1 to 30 cm/s. Velocities above this 14 cm/s were excluded from figures because not all mice ran faster than 14 cm/s. For each lap, the ROIs average fluorescence in each velocity bin was calculated and then averaged across all laps in each environment to find the velocity binned activity in the familiar and novel environments.

Motion initiation aligned fluorescence

Periods where mice were immobile (velocity < 5 cm/s) for at least 1.5s then proceed to run (velocity > 5cm/s) for at least 3s were identified. The fluorescent activity for ROIs for these periods was aligned to the frame mice began running (velocity crossed above 5cm/s). The average aligned fluorescent activity of each ROI was then determined for each environment.

Linear regression analysis

To assess dynamics between each of the above measures and calcium activity of LC and VTA axons, we performed linear regression on the population’s familiar environment data and significance was assessed with an F test. To compare the dynamics between LC and VTA axons, we performed exact testing based on Monte-Carlo resampling (1000 resamples with sample size matching the lower sample size condition) as detailed in legends (**Fig. 1E** [↗](#)).

To assess the changing position and velocity encoding of LC axons following exposure to a novel environment, we performed linear regression on the population fluorescence data of the average of the last 4 laps in the familiar environment, and each of the first three laps in the novel environment for each measure. The significance for the fit of each line was assessed with an F test, and an ANCOVA was conducted to test for differences in slope between the four laps. The same process was conducted for the motion initiation dynamics, but only using ROIs in mice who paused within the first 2 laps and 30s following exposure to the novel environment.

Novel response analysis

To examine the response of LC and VTA axons to the novel VR environment, the fluorescence data was normalized by the mean for each ROIs and aligned to the frame where the mice were switched to the novel environment and the mean normalized F for LC and VTA ROIs at each time point was calculated. Baseline fluorescent activity was then calculated for LC and VTA ROIs separately by generating 1000 shuned traces of the ROIs calcium activity and subsampling down to the sample size (90 for LC; 7 for VTA) 1000 times and finding the mean of the subsampled shunes. The mean and 95% CI of all 1000 subsamples was found and the mean activity of LC and VTA ROIs was considered significantly elevated when it passed above the 95% CI of the shuned data. The same process was repeated to define a baseline for the time binned data (fluorescent activity divided into 50 frame bins) and the lap binned data (mean activity for each lap).

Additionally, to account for changes in behavior between the familiar and novel environments, periods where the animals were immobile (velocity ≤ 0.2 cm/s) were removed and running periods were concatenated together and aligned to the switch to the novel environment. Here, we again defined a baseline for the time mean traces, time binned activity, and the lap binned activity using the above bootstrapping approach.

Figure graphics

All figure graphics including **Fig. 1A-B** and **Fig. 2A** were created using BioRender.com.

Code availability

Scripts used for data analysis are available on Github https://github.com/chadheer/LC_VTA_paper

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Author contributions

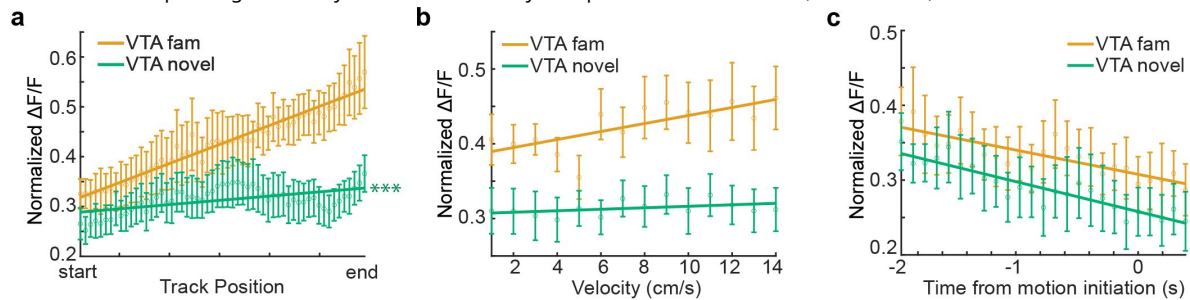
C.H. and M.S. conceived and designed the experiments. C.H. collected and analyzed the data. C.H. and M.S. interpreted the data and wrote the manuscript.

Declaration of interests

The authors declare no competing interests.

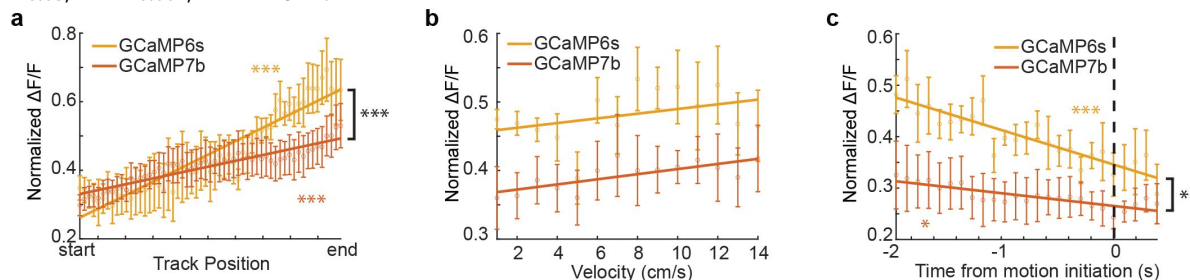
Supplementary Figure 1.

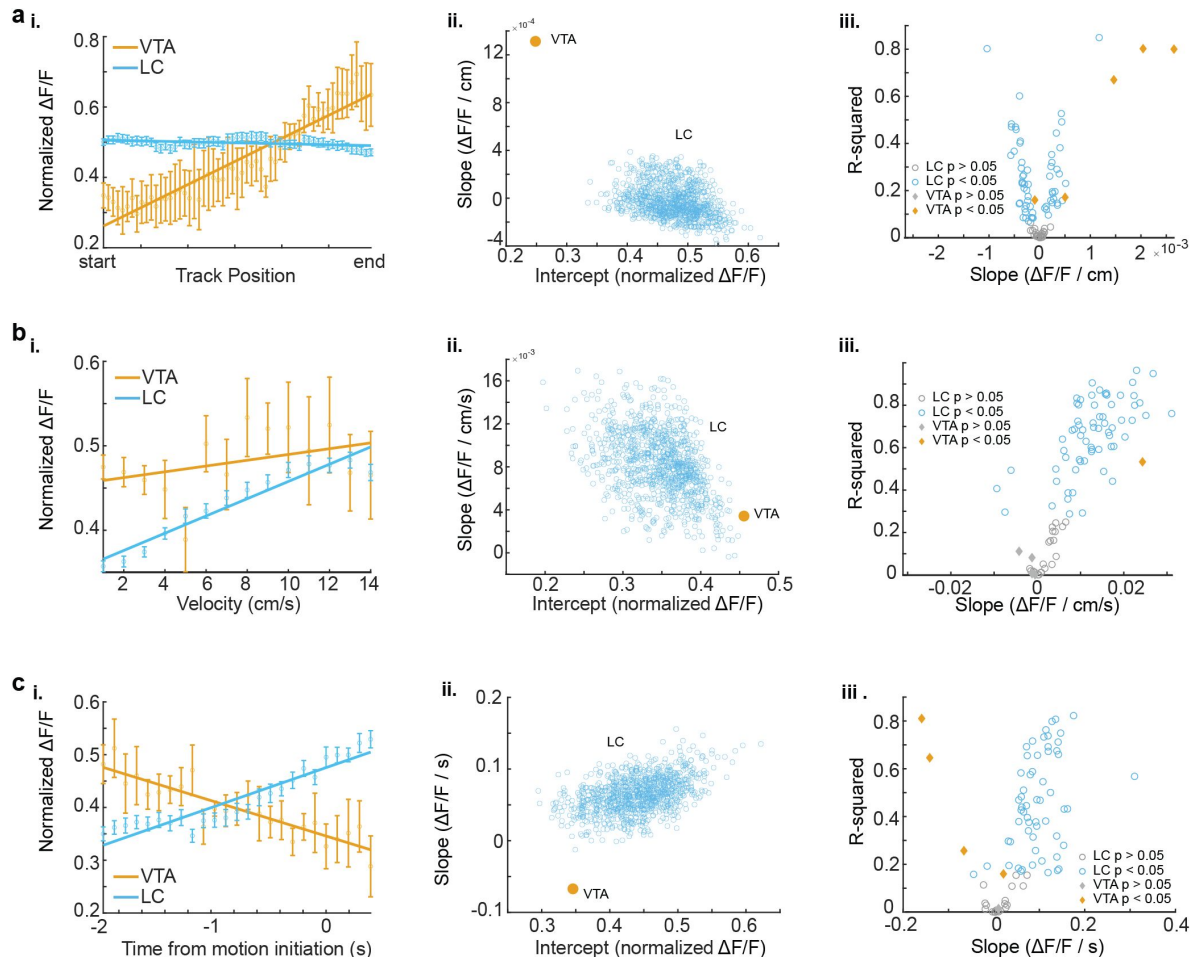
a,i, Population average position binned $\Delta F / F \pm$ s.e.m. of VTA ROIs ($n = 7$ ROIs in 7 mice) in the familiar (orange) and novel (green) rewarded environments. Linear regression, F test, Rewarded, $P < 1e - 21$, Unrewarded, $P < 0.01$. ii, Same data as (a,i) binned by velocity. Linear regression, F test, Rewarded, $P < 0.05$, Unrewarded, $P = 0.57$. iii, Same data as (a,i) aligned to motion onset. Linear regression, F test, Rewarded, $P < 0.01$, Unrewarded, $P < 0.001$. The slope of each novel measure was compared to the familiar laps using a one-way ANCOVA with Tukey HSD post hoc test. * $P < 0.05$, ** $P < 0.001$, *** $P < 1e - 4$.



Supplementary Figure 2.

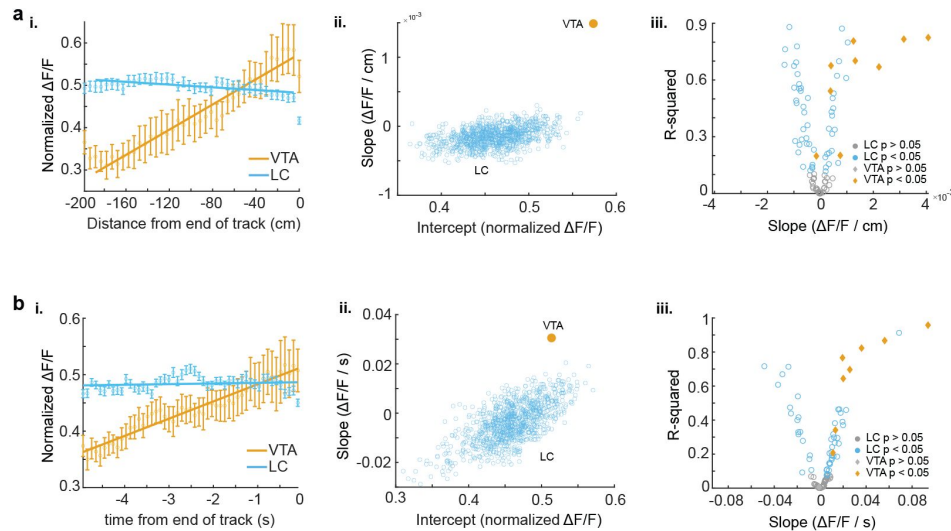
a, Population average position binned $\Delta F / F \pm$ s.e.m. of VTA GCaMP6s ROIs (orange, $n = 5$ ROIs in 4 mice) and VTA GCaMP7b ROIs (dark orange, $n = 4$ ROIs in 4 mice) in the familiar environment. Linear regression, F test, GCaMP6s, $P = 2.4264e - 32$, GCaMP7b, $P = 1.9576e - 14$. b, Same data as (a) binned by velocity. Linear regression, F test, GCaMP6s, $P = 0.21076$, GCaMP7b, $P = 0.14113$. c, Same data as (a) aligned to motion onset. Linear regression, F test, GCaMP6s, $P = 2.0974e - 8$, GCaMP7b, $P = 0.026893$. The slopes for the two GCaMP variants were compared using a one-way ANCOVA with Tukey HSD post hoc test * $P < 0.05$, ** $P < 0.001$, *** $P < 1e - 4$.





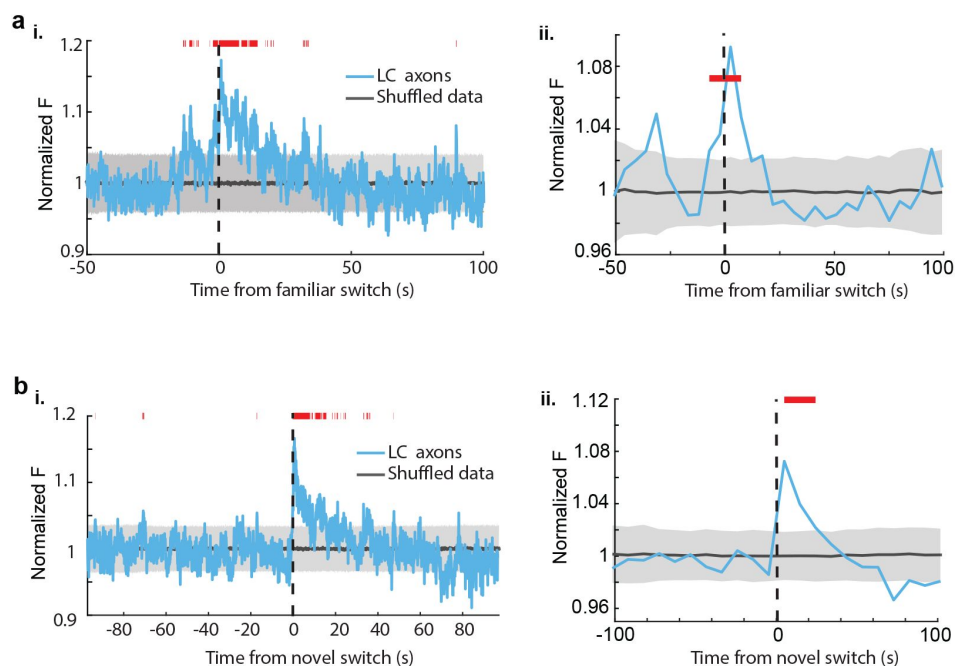
Supplementary Figure 3.

a,i, Population average position binned $\Delta F/F \pm$ s.e.m. in axon-GCaMP6s expressing VTA ROIs (orange, $n = 5$ ROIs in 4 mice) and LC ROIs (blue, $n = 87$ ROIs from 27 sessions in 17 mice). Linear regression analysis (on all data points, not means) shows that the population of VTA ROIs increase activity during approach of the end of the track while the population of LC ROIs have consistent activity throughout all positions. Linear regression, F test, VTA, $P = 2.42e - 32$, LC, $P = 0.00396$. ii, The LC data set was resampled 1000x using $n = 5$ axons to match the number of VTA ROIs and the slope and intercept of the regression line were measured each time (blue dots). The VTA slope is steeper than all LC slopes indicating a stronger positive relationship between position and activity for VTA inputs. iii, Linear regression of position binned activity of individual VTA (orange diamonds), and LC (blue, circles) axons. The majority (4/5) of VTA axons show a significant positive relationship with position while LC axons show both a positive (25/87) and negative (37/87) relationship. b, i, Same data as (a, i), binned by velocity. Linear regression shows that the population of LC ROIs have a significant relationship with velocity. Linear regression, F test, VTA, $P = 0.211$, LC, $P < 1e - 68$. ii, Resampling shows the VTA slope is within the resampled LC slopes indicating similar relationships with velocity. iii, Linear regression of individual VTA and LC axons shows the majority (63/87) of LC axons have a significant positive relationship with velocity while only 1 VTA axon shows this relationship. c, i Same data as (a, i), aligned to motion onset. Linear regression shows that the population of VTA axons have a negative slope prior to motion onset while LC axons have positive slope. Linear regression, F test, VTA, $P = 2.10e - 081$, LC, $P = 5.51e - 66$. ii, Resampling shows the VTA slope is negative while all resampled LC slopes are positive. iii, Linear regression of individual VTA and LC axons shows the majority (56/87) of LC axons have a significant positive slope prior to motion onset while the majority (3/5) of VTA axons have a negative slope.



Supplementary Figure 4.

a,i, Population activity $\Delta F/F \pm$ s.e.m. binned by the virtual distance to reward for VTA ROIs (orange, 200m track $n = 9$ ROIs in 8 mice) and LC ROIs (blue, 300m track, $n = 87$ ROIs from 27 sessions in 17 mice) in the familiar environment. Linear regression, F test, VTA, $P = 1.83e - 28$, LC, $P = 8.77e - 06$. ii, The LC data set was resampled 1000x using $n = 9$ axons to match the number of VTA Rois and the slope and intercept of the regression line were measured each time (blue dots). The VTA slope is steeper than all LC slopes indicating a stronger positive relationship between position and activity for VTA inputs.iii, Linear regression of position binned activity of individual VTA (orange diamonds), and LC (blue, circles) axons. The majority (8/9) of VTA axons show a significant positive relationship with position while LC axons show both a positive (21/87 axons from 9 sessions in 8 mice) and negative (32/87 axons from 15 sessions in 9 mice) relationship.b, i, Same data as (a, i,) averaged by time to reward. Linear regression shows that the population of VTA axons has a significant positive relationship with time to reward. Linear regression, F test, VTA, $P = 4.44e - 25$, LC, $P = 0.119$. ii, Resampling shows the VTA slope is above the resampled LC slopes indicating VTA ROIs have a stronger positive relationship with time to reward. iii, Linear regression of individual VTA and LC axons shows the majority (8/9) of VTA axons have a significant positive relationship with time to reward while LC axons show both a significant positive (31/87 axons from 14 sessions in 11 mice) and negative (17/87 axons from 7 sessions in 4 mice) relationship



Supplementary Figure 5.

a, i, Mean normalized fluorescence of LC ROIs ($n = 50$, 11 sessions in 9 mice) aligned to the switch from dark to the familiar environment. To define a baseline and 95% CI (gray shaded region), 1000 shuffles were created from the calcium traces and down sampled to match the sample size and averaged. This was repeated 1000 times and the mean and 95% CI of this shuffled data was determined for each frame. Red lines indicate periods where two or more consecutive frames passed above the 95% CI of the shuffled baseline. **a,ii**, The normalized fluorescence of all LC binned into 50 frame bins. The baseline and 95% CI (gray shaded region) was defined using the same method as in (a). Red lines indicate 2 or more consecutive bins above the baseline 95% CI. **b, i**, Mean normalized fluorescence of LC ROIs ($n = 87$) during immobile periods aligned to the switch from dark to the familiar environment. The baseline and 95% CI (gray shaded region) was defined using the same method as in (a). Red lines indicate 2 or more consecutive bins above the baseline 95% CI. **b, ii** The normalized fluorescence of LC axons during immobile periods binned into 50 frame bins. The baseline and 95% CI (gray shaded region) was defined using the same method as in (a). Red lines indicate 2 or more consecutive bins above the baseline 95% CI.

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Reviewer #1 (Public Review):

Summary:

Heer and Sheffield used 2 photon imaging to dissect the functional contributions of convergent dopamine and noradrenaline inputs to the dorsal hippocampus CA1 in head restrained mice running down a virtual linear path. Mice were trained to collect water reward at the end of the track and on test days, calcium activity was recorded from dopamine (DA) axons originating in ventral tegmental area (VTA, n=7) and noradrenaline axons from

the locus coeruleus (LC, $n=87$) under several conditions. When mice ran laps in a familiar environment, VTA DA axons exhibited ramping activity along the track that correlated with distance to reward and velocity to some extent, while LC input activity remained constant across the track, but correlated invariantly with velocity and time to motion onset. A subset of recordings taken when the reward was removed showed diminished ramping activity in VTA DA axons, but no changes in the LC axons, confirming that DA axon activity is locked to reward availability. When mice were subsequently introduced to a new environment, the ramping to reward activity in the DA axons disappeared, while LC axons showed a dramatic increase in activity lasting 90s (6 laps) following the environment switch. In the final analysis, the authors sought to disentangle LC axon activity induced by novelty vs. behavioral changes induced by novelty by removing periods in which animals were immobile and established that the activity observed in the first 2 laps reflected novelty-induced signal in LC axons.

The revised manuscript included additional evidence of increased (but transient) signal in LC axons after a transition to a novel environment during periods of immobility, and also that a change from dark to familiar environment induces a peak in LC axon activity, showing that LC input to dCA1 may not solely signal novelty.

Strengths:

The results presented in this manuscript provide insights into the specific contributions of catecholaminergic input to the dorsal hippocampus CA1 during spatial navigation in a rewarded virtual environment, offering a detailed analysis at the resolution of single axons. The data analysis is thorough and possible confounding variables and data interpretation are carefully considered.

The authors have addressed my concerns in a thorough manner. The reviewer also appreciates the increased transparency of reporting in the revised manuscript.

Weaknesses:

Listed below are some remaining comments.

The increase in LC activity with any change in environment (from familiar to novel or from dark to familiar) suggests that LC input acts not solely as a novelty signal, but as a general arousal or salience signal in response to environmental changes. Based on this, I have a couple of questions:

- Is the overall claim that LC input to the dHC signals novelty still valid based on observed findings - as claimed throughout the manuscript?
- Would the omission of a reward be considered a salient change in the environment that activates LC signals, or is the LC not involved with processing reward-related information? Has the activity of LC and VTA axons been analysed in the seconds following reward presentation and/or omission?

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Reviewer #2 (Public Review):

Summary:

The authors used 2-photon Ca^{2+} -imaging to study the activity of ventral tegmental area (VTA) and locus coeruleus (LC) axons in the CA1 region of the dorsal hippocampus in head-fixed male mice moving on linear paths in virtual reality (VR) environments.

The main findings were as follows:

- In a familiar environment, activity of both VTA axons and LC axons increased with the

mice's running speed on the Styrofoam wheel, with which they could move along a linear track through a VR environment.

- VTA, but not LC, axons showed marked reward position-related activity, showing a ramping-up of activity when mice approached a learned reward position.
- In contrast, activity of LC axons ramped up before initiation of movement on the Styrofoam wheel.
- In addition, exposure to a novel VR environment increased LC axon activity, but not VTA axon activity.

Overall, the study shows that the activity of catecholaminergic axons from VTA and LC to dorsal hippocampal CA1 can partly reflect distinct environmental, behavioral and cognitive factors. Whereas both VTA and LC activity reflected running speed, VTA, but not LC axon activity reflected the approach of a learned reward and LC, but not VTA, axon activity reflected initiation of running and novelty of the VR environment.

I have no specific expertise with respect to 2-photon imaging, so cannot evaluate the validity of the specific methods used to collect and analyse 2-photon calcium imaging data of axonal activity.

Strengths:

- (1) Using a state-of-the-art approach to record separately the activity of VTA and LC axons with high temporal resolution in awake mice moving through virtual environments, the authors provide convincing evidence that activity of VTA and LC axons projecting to dorsal CA1 reflect partly distinct environmental, behavioral and cognitive factors.
- (2) The study will help a) to interpret previous findings on how hippocampal dopamine and norepinephrine or selective manipulations of hippocampal LC or VTA inputs modulate behavior and b) to generate specific hypotheses on the impact of selective manipulations of hippocampal LC or VTA inputs on behavior.

Weaknesses:

- (1) The findings are correlational and do not allow strong conclusions on how VTA or LC inputs to dorsal CA1 affect cognition and behavior. However, as indicated above under Strengths, the findings will aid the interpretation of previous findings and help to generate new hypotheses as to how VTA or LC inputs to dorsal CA1 affect distinct cognitive and behavioral functions.

- (2) Some aspects of the methodology would benefit from clarification.

First, to help others to better scrutinize, evaluate and potentially to reproduce the research, the authors may wish to check if their reporting follows the ARRIVE (Animal Research: Reporting of In Vivo Experiments) guidelines for the full and transparent reporting of research involving animals (<https://arriveguidelines.org/>). For example, I think it would be important to include a sample size justification (e.g., based on previous studies, considerations of statistical power, practical considerations or a combination of these factors). The authors should also include the provenance of the mice. Moreover, although I am not an expert in 2-photon imaging, I think it would be useful to provide a clearer description of exclusion criteria for imaging data (see below, Recommendations for the authors).

Second, why were different linear tracks used for studies of VTA and LC axon activity (from line 362)? Could this potentially contribute to the partly distinct activity correlates that were found for VTA and LC axons?

Third, the authors seem to have used two different criteria for defining immobility.

Immobility was defined as moving at <5 cm/s for the behavioral analysis in Fig. 3a, but as <0.2 cm/s for the imaging data analysis in Fig. 4 (see legends to these figures and also see Methods,

from line 447, line 469, line 498)? I do not understand why, and it would be good if the authors explained this.

(3) In the Results section (from line 182) the authors convincingly addressed the possibility that less time spent immobile in the novel environment may have contributed to the novelty-induced increase of LC axon activity in dorsal CA1 (Fig. 4). In addition, initially (for the first 2-4 laps), the mice also ran more slowly in the novel environment (Fig. 3aIII, top panel). Given that LC and VTA axon activity were both increasing with velocity (Fig. 1F), reduced velocity in the novel environment may have reduced LC and VTA axon activity, but this possibility was not addressed. Reduced LC axon activity in the novel environment could have blunted the novelty-induced increase. More importantly, any potential novelty-induced increase in VTA axon activity could have been masked by decreases in VTA axon activity due to reduced velocity. The latter may help to explain the discrepancy between the present study and previous findings that VTA neuron firing was increased by novelty (see Discussion, from line 243). It may be useful for the authors to address these possibilities based on their data in the Results section, or to consider them in their Discussion.

(4) Sensory properties of the water reward, which the mice may be able to detect, could account for reward-related activity of VTA axons (instead of an expectation of reward). Do the authors have evidence that this is not the case? Occasional probe trials, intermixed with rewarded trials, could be used to test for this possibility.

REVIEW OF THE REVISED MANUSCRIPT

I thank the authors for their responses addressing some of the weaknesses I raised in my original comments.

Regarding their clarification of some methodological issues [Point 2) above], I have a few additional comments:

- I appreciate that the authors clearly state the sample sizes contributing to the data. However, sample size justifications (e.g. based on previous studies, considerations of statistical power, practical considerations or a combination of these factors) are still lacking.
- It is good that the authors have now clearly indicated how many mice they excluded due to lack of GCaMP expression or due to failure to reach the behavioral criteria. They also indicated that they discarded some of the collected datasets, based on the visual assessment of imaging sessions and the registration metrics output by suite2p. I appreciate that this may be common practice (although I am not using 2-photon imaging myself). However, I note that to minimize the risk of experimenter bias and improve reproducibility, it would be preferable to have more clearly defined quantitative criteria for such exclusions.
- The authors clarified in their response why they used two different linear tracks for their studies of VTA and LC axon activity. I would encourage them to include this clarification in the manuscript. From the authors' response, I understand that they chose the different track lengths to facilitate comparison to previous studies involving LC and VTA axon recordings. However, given that the present paper aimed to compare LC and VTA axon recordings, the use of different track lengths for LC and VTA axon recordings remains a limitation of the present paper.

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Reviewer #3 (Public Review):

Summary:

Heer and Sheffield provide a well-written manuscript that clearly articulates the theoretical motivation to investigate specific catecholaminergic projections to dorsal CA1 of the hippocampus during a reward-based behavior. Using 2-photon calcium imaging in two

groups of cre transgenic mice, the authors examine activity of VTA-CA1 dopamine and LC-CA1 noradrenergic axons during reward seeking in a linear track virtual reality (VR) task. The authors provide a descriptive account of VTA and LC activities during walking, approach to reward, and environment change. Their results demonstrate LC-CA1 axons are activated by walking onset, modulated by walking velocity, and heighten their activity during environment change. In contrast, VTA-CA1 axons were most activated during approach to reward locations. Together the authors provide a functional dissociation between these catecholamine projections to CA1. A major strength to their approach is the methodological rigor of 2-photon recording, data processing, and analysis approaches to accommodate their unequal LC-CA1 and VTA-CA1 sample sizes. These important systems neuroscience studies provide solid evidence that will contribute to the broader field of navigation and memory.

Weaknesses:

The conclusions of this manuscript are mostly well supported by the data. However, increasing the sample size of the VTA-CA1 group and using experimental methods that are identical among LC-CA1 and VTA-CA1 groups would help to fully support the author's conclusions.

<https://doi.org/10.7554/eLife.95213.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

Heer and Sheffield used 2 photon imaging to dissect the functional contributions of convergent dopamine and noradrenaline inputs to the dorsal hippocampus CA1 in head-restrained mice running down a virtual linear path. Mice were trained to collect water rewards at the end of the track and on test days, calcium activity was recorded from dopamine (DA) axons originating in the ventral tegmental area (VTA, n=7) and noradrenaline axons from the locus coeruleus (LC, n=87) under several conditions. When mice ran laps in a familiar environment, VTA DA axons exhibited ramping activity along the track that correlated with distance to reward and velocity to some extent, while LC input activity remained constant across the track, but correlated invariantly with velocity and time to motion onset. A subset of recordings taken when the reward was removed showed diminished ramping activity in VTA DA axons, but no changes in the LC axons, confirming that DA axon activity is locked to reward availability. When mice were subsequently introduced to a new environment, the ramping to reward activity in the DA axons disappeared, while LC axons showed a dramatic increase in activity lasting 90 s (6 laps) following the environment switch. In the final analysis, the authors sought to disentangle LC axon activity induced by novelty vs. behavioral changes induced by novelty by removing periods in which animals were immobile and established that the activity observed in the first 2 laps reflected novelty-induced signal in LC axons.

Strengths:

The results presented in this manuscript provide insights into the specific contributions of catecholaminergic input to the dorsal hippocampus CA1 during spatial navigation in a rewarded virtual environment, offering a detailed analysis of the resolution of single

axons. The data analysis is thorough and possible confounding variables and data interpretation are carefully considered.

Weaknesses:

Aspects of the methodology, data analysis, and interpretation diminish the overall significance of the findings, as detailed below.

The LC axonal recordings are well-powered, but the DA axonal recordings are severely underpowered, with recordings taken from a mere 7 axons (compared to 87 LC axons).

Additionally, 2 different calcium indicators with differential kinetics and sensitivity to calcium changes (GCaMP6S and GCaMP7b) were used (n=3, n=4 respectively) and the data pooled. This makes it very challenging to draw any valid conclusions from the data, particularly in the novelty experiment. The surprising lack of novelty-induced DA axon activity may be a false negative. Indeed, at least 1 axon (axon 2) appears to be showing a novelty-induced rise in activity in Figure 3C. Changes in activity in 4/7 axons are also referred to as a 'majority' occurrence in the manuscript, which again is not an accurate representation of the observed data.

We appreciate the reviewer's detailed feedback regarding the analysis of VTA axons in our dataset. The relatively low sample size for VTA axons is due to their sparsity in the dCA1 region of the hippocampus and the inherent difficulty in recording from these axons. VTA axons are challenging to capture due to their low baseline fluorescence and long-range axon segments, resulting in a typical yield of only a single axon per field of view (FOV) per animal. In contrast, LC axons are more abundant in dCA1.

To address the disparity in sample sizes between LC and VTA axons, we down-sampled the LC axons to match the number of VTA axons, repeating this process 1000 times to create a distribution. However, we acknowledge the reviewer's concern that the relatively low sample size for VTA axons might result in insufficient sampling of this population. Increasing the baseline expression of GCaMP to record from VTA axons requires several months, limiting our ability to quickly expand the sample size.

In response to the reviewer's comments, we have added recordings from 2 additional VTA axons, increasing the sample size from 7 to 9. We re-analyzed all data from the familiar environment with n=9 VTA axons, comparing them to down-sampled LC axons as previously described. However, the additional axons were not recorded in the novel environment. We agree with the reviewer that the lack of novelty-induced DA axon activity may be a false negative. To address this, we have revised the description of our results to include the following sentence:

"However, 1 VTA ROI showed an increase in activity immediately following exposure to novelty, indicating heterogeneity across VTA axons in CA1, and the lack of a novelty signal on average may be due to a small sample size."

Regarding the use of two different GCaMP constructs, we understand the reviewer's concern. We used GCaMP6s and GCaMP7b variants to determine if one would improve the success rate of recording from VTA axons. Given the long duration of these experiments and the low yield, we pooled the data from both GCaMP variants to increase statistical power. However, we recognize the importance of verifying that there are no differences in the signals recorded with these variants.

With the addition of 2 VTA DA axons expressing GCaMP6s, we now have n=5 GCaMP6s and n=4 GCaMP7b VTA DA axons. This allowed us to compare the activity of the two sensors in the familiar environment. As shown in new Supplementary Figure 2, both sets of axons responded similarly to the variables measured: position in VR, time to motion onset, and

animal velocity (although the GCaMP6s expressing axons showed stronger correlations). Since all LC axons recorded expressed GCaMP6s, we also specifically compared VTA GCaMP6s axons to LC GCaMP6s axons (Supp Fig. 3). Our conclusions remained consistent when comparing this subset of VTA axons to LC axons.

Overall, our paper now includes comparisons of combined VTA axons (n=9) and separately the GCaMP6s-expressing VTA axons (n=5) with LC axons. Both datasets support our initial conclusions that VTA axons signal proximity to reward, while LC axons encode velocity and motion initiation in familiar environments.

The authors conducted analysis on recording data exclusively from periods of running in the novelty experiment to isolate the effects of novelty from novelty-induced changes in behavior. However, if the goal is to distinguish between changes in locus coeruleus (LC) axon activity induced by novelty and those induced by motion, analyzing LC axon activity during periods of immobility would enhance the robustness of the results.

We appreciate the reviewer's insightful suggestion to analyze LC axon activity during periods of immobility to distinguish between changes induced by novelty and those induced by motion. This additional analysis would indeed strengthen our conclusions regarding the LC novelty signal.

In response to this suggestion, we performed the same analysis as before, but focused on periods of immobility. Our findings indicate that following exposure to novelty, there was a significant increase in LC activity specifically during immobility. This supports the idea that LC axons produce a novelty signal that is independent of novelty-induced behavioral changes. The results of this analysis are now presented in new Supplementary Figure 5b

The authors attribute the ramping activity of the DA axons to the encoding of the animals' position relative to reward. However, given the extensive data implicating the dorsal CA1 in timing, and the remarkable periodicity of the behavior, the fact that DA axons could be signalling temporal information should be considered.

This is an insightful comment regarding the potential role of VTA DA axons in signaling temporal information. We agree that VTA DA axons could indeed be encoding temporal information, as previous work from our lab has shown that these axons exhibit ramping activity when averaged by time to reward (Krishnan et al., 2022).

To address this, we have now examined DA axon activity relative to time to reward, as shown in new Supplementary Figure 4. Our analysis confirms that these axons ramp up in activity relative to time to reward. Given the periodicity of our mice's behavior in these experiments, as the reviewer correctly points out, we are unable to distinguish between spatial proximity to reward and time to reward. We have added a sentence to our paper highlighting this limitation and stating that further experiments are necessary to differentiate these two variables.

Krishnan, L.S., Heer, C., Cherian, C., Sheffield, M.E. Reward expectation extinction restructures and degrades CA1 spatial maps through loss of a dopaminergic reward proximity signal. *Nat Commun* 13, 6662 (2022).

The authors should explain and justify the use of a longer linear track (3m, as opposed to 2m in the DAT-cre mice) in the LC axon recording experiments.

We appreciate the reviewer's insightful comment regarding the use of a longer linear track (3m, as opposed to 2m in the DAT-cre mice) in the LC axon recording experiments. The choice of a 3m track for LC axon recordings was made to align with a previous experiment from our

lab (Dong et al., 2021), in which mice were exposed to a novel 3m track while CA1 pyramidal cell populations were recorded. In that study, we detailed the time course of place field formation within the novel track. Our current hypothesis is that LC axons signal novelty, and we aimed to investigate whether the time course of LC axon activity aligns with the time course of place field formation. This hypothesis, and the potential role of LC axons in facilitating plasticity for new place field formation, is further discussed in the Discussion section of our paper.

For the VTA axon recordings, we utilized a 2m track, consistent with another recent study from our lab (Krishnan et al., 2022), where reward expectation was manipulated, and CA1 pyramidal cell populations were recorded. By matching the track length to this prior study, we aimed to explore how VTA dopaminergic inputs to CA1 might influence CA1 population dynamics along the track under conditions of varying reward expectations.

We acknowledge that using different track lengths for LC and VTA recordings introduces a variable that could potentially confound direct comparisons. To address this, we normalized the track lengths for our LC versus VTA comparison analysis. This normalization allowed us to directly compare patterns of activity across the two types of axons by adjusting the data to a common scale, thereby ensuring that any observed differences or similarities are attributable to the intrinsic properties of the axons rather than differences in track lengths. By doing so, we could assess relative changes in activity levels at matched spatial bins.

Although the experiences of the animals on the different track lengths are not identical, our observations suggest that LC and VTA axon signals are not majorly influenced by variations in track length. LC axons are associated with velocity and a pre-motion initiation signal, neither of which are affected by track length. VTA axons, which also correlate with velocity, can be compared to LC axon velocity signals because mice reach maximal velocity very quickly along the track, well before the end of the 2m track. The range of velocities are therefore captured on both track lengths. While VTA axons exhibit ramping activity as they approach the reward zone—a signal potentially modulated by track length—LC axons do not show such ramping to reward signals. Thus, a comparison across different track lengths is justified for this aspect of our analysis.

To further enhance the rigor of our comparisons between axon dynamics recorded on 2m and 3m tracks, we conducted an additional analysis plotting axon activity by time to reward and actual (un-normalized) distance from reward (Supplementary Figure 4). This analysis revealed very similar signals between the two sets of axons, supporting our initial conclusions.

We thank the reviewer for raising this important point and hope that our detailed explanation and additional analysis address their concern.

Krishnan, L.S., Heer, C., Cherian, C., Sheffield, M.E. Reward expectation extinction restructures and degrades CA1 spatial maps through loss of a dopaminergic reward proximity signal. *Nat Commun* 13, 6662 (2022).

Dong, C., Madar, A. D. & Sheffield, M.E. Distinct place cell dynamics in CA1 and CA3 encode experience in new environments. *Nat Commun* 12, 2977 (2021).

Reviewer #2 (Public Review):

Summary:

The authors used 2-photon Ca2+-imaging to study the activity of ventral tegmental area (VTA) and locus coeruleus (LC) axons in the CA1 region of the dorsal hippocampus in head-fixed male mice moving on linear paths in virtual reality (VR) environments.

The main findings were as follows:

- In a familiar environment, the activity of both VTA axons and LC axons increased with the mice's running speed on the Styrofoam wheel, with which they could move along a linear track through a VR environment.

- VTA, but not LC, axons showed marked reward position-related activity, showing a ramping-up of activity when mice approached a learned reward position.

- In contrast, the activity of LC axons ramped up before the initiation of movement on the Styrofoam wheel.

- In addition, exposure to a novel VR environment increased LC axon activity, but not VTA axon activity.

Overall, the study shows that the activity of catecholaminergic axons from VTA and LC to dorsal hippocampal CA1 can partly reflect distinct environmental, behavioral, and cognitive factors. Whereas both VTA and LC activity reflected running speed, VTA, but not LC axon activity reflected the approach of a learned reward, and LC, but not VTA, axon activity reflected initiation of running and novelty of the VR environment.

I have no specific expertise with respect to 2-photon imaging, so cannot evaluate the validity of the specific methods used to collect and analyse 2-photon calcium imaging data of axonal activity.

Strengths:

(1) Using a state-of-the-art approach to record separately the activity of VTA and LC axons with high temporal resolution in awake mice moving through virtual environments, the authors provide convincing evidence that the activity of VTA and LC axons projecting to dorsal CA1 reflect partly distinct environmental, behavioral and cognitive factors.

(2) The study will help a) to interpret previous findings on how hippocampal dopamine and norepinephrine or selective manipulations of hippocampal LC or VTA inputs modulate behavior and b) to generate specific hypotheses on the impact of selective manipulations of hippocampal LC or VTA inputs on behavior.

Weaknesses:

(1) The findings are correlational and do not allow strong conclusions on how VTA or LC inputs to dorsal CA1 affect cognition and behavior. However, as indicated above under Strengths, the findings will aid the interpretation of previous findings and help to generate new hypotheses as to how VTA or LC inputs to dorsal CA1 affect distinct cognitive and behavioral functions.

(2) Some aspects of the methodology would benefit from clarification.

First, to help others to better scrutinize, evaluate, and potentially to reproduce the research, the authors may wish to check if their reporting follows the ARRIVE (Animal Research: Reporting of In Vivo Experiments) guidelines for the full and transparent reporting of research involving animals (<https://arriveguidelines.org/>). For example, I think it would be important to include a sample size justification (e.g., based on previous studies, considerations of statistical power, practical considerations, or a combination of these factors). The authors should also include the provenance of the mice. Moreover, although I am not an expert in 2-photon imaging, I think it would be useful to provide a clearer description of exclusion criteria for imaging data.

We thank the reviewer for helping us formalize the scientific rigor of our study. There are ten ARRIVE Guidelines and we have addressed most of them in our study already. However, there is an opportunity to add detail. We have listed below all ten points and how we have addressed each one (and point out any new additions):

(1) Experimental design - we go into great depth explaining the experimental set-up, how we used the autofluorescent blebs as imaging controls, how we controlled for different sample sizes between the two populations, and the statistical tests used for comparisons. We also carefully accounted for animal behavior when quantifying and describing axon dynamics both in the familiar and novel environments.

(2) Sample size - we state both the number of ROIs and mice for each analysis. We have now also added the number of mice we observed specific types of activity in.

(3) Inclusion/exclusion criteria - The following has now been added to the Methods section: Out of the 36 NET-Cre mice injected, 15 were never recorded from for either failing to reach behavioral criteria, or a lack of visible expression in axons. Out of the 54 DAT-Cre mice injected, imaging was never conducted in 36 of them for lack of expression or failing to reach behavioral criteria. Out of the remaining 21 NET-CRE, 5 were excluded for heat bubbles, z-drift, or bleaching, while 10 DAT-Cre were excluded for the same reasons. This was determined by visually assessing imaging sessions, followed by using the registration metrics output by suite2p. This registration metric conducted a PCA on the motion-corrected ROIs and plotted the first PC. If the PC drifted largely, to the point where no activity was apparent, the video was excluded from analysis.

(4) Randomization - Already included in the paper is a description of random downsampling of LC axons to make statistical comparisons with VTA axons. LC axons were selected pseudo-randomly (only one axon per imaging session) to match VTA sampling statistics. This randomization was repeated 1000 times and comparisons were made against this random distribution.

(5) Blinding-masking - no blinding/masking was conducted as no treatments were given that would require this. We will include this statement in the next version.

(6) Outcomes - We defined all outcomes measured, such as those related to animal behavior and axon signaling.

(7) Statistical methods - None of the reviewers had any issues regarding our description of statistical methods, which we described in great detail in this version of the paper.

(8) Experimental animals - We have now described that DAT- Cre mice were obtained through JAX labs, and NET-Cre mice were obtained from the Tonegawa lab (Wagatsuma et al. 2017). This was absent in the initial version of the paper.

(9) Experimental procedure - Already listed in great detail in Methods section.

(10) Results - Rigorously described in detail for behaviors and related axon dynamics.

Wagatsuma, Akiko, Teruhiro Okuyama, Chen Sun, Lillian M. Smith, Kuniya Abe, and Susumu Tonegawa. "Locus Coeruleus Input to Hippocampal CA3 Drives Single-Trial Learning of a Novel Context." *Proceedings of the National Academy of Sciences* 115, no. 2 (January 9, 2018): E310–16. <https://doi.org/10.1073/pnas.1714082115>.

Second, why were different linear tracks used for studies of VTA and LC axon activity (from line 362)? Could this potentially contribute to the partly distinct activity correlates that were found for VTA and LC axons?

We thank the reviewer for pointing this out and giving us a chance to address it directly. A detailed response to this is written above for a similar comment from reviewer 1.

Third, the authors seem to have used two different criteria for defining immobility. Immobility was defined as moving at <5 cm/s for the behavioral analysis in Figure 3a, but as <0.2 cm/s for the imaging data analysis in Figure 4 (see legends to these figures and also see Methods, from line 447, line 469, line 498)? I do not understand why, and it would be good if the authors explained this.

This is a typo leftover from before we converted velocity from rotational units of the treadmill to cm/s. This has now been corrected.

(3) In the Results section (from line 182) the authors convincingly addressed the possibility that less time spent immobile in the novel environment may have contributed to the novelty-induced increase of LC axon activity in dorsal CA1 (Figure 4). In addition, initially (for the first 2-4 laps), the mice also ran more slowly in the novel environment (Figure 3aIII, top panel). Given that LC and VTA axon activity were both increasing with velocity (Figure 1F), reduced velocity in the novel environment may have reduced LC and VTA axon activity, but this possibility was not addressed. Reduced LC axon activity in the novel environment could have blunted the novelty-induced increase. More importantly, any potential novelty-induced increase in VTA axon activity could have been masked by decreases in VTA axon activity due to reduced velocity. The latter may help to explain the discrepancy between the present study and previous findings that VTA neuron firing was increased by novelty (see Discussion, from line 243). It may be useful for the authors to address these possibilities based on their data in the Results section, or to consider them in their Discussion.

We appreciate the reviewer's insightful comment regarding the potential impact of decreased velocity on novelty responses in LC and VTA axons. The decreased velocity in the novel environment could lead to a diminished novelty response in LC axons and could mask a subtle novelty signal in VTA axons. We have now included the following points in our discussion:

“In addition, as noted above, on average we did observe a velocity associated signal in VTA axons. When mice were exposed to the novel environment their velocity initially decreased. This would be expected to reduce the average signal across the VTA axon population relative to the higher velocity in the familiar environment. It is possible that this decrease could somewhat mask a subtle novelty induced signal in VTA axons. Therefore, additional experiments should be conducted to investigate the heterogeneity of these axons and their activity under different experimental conditions during tightly controlled behavior.”

“As discussed above, the slowing down of animal behavior in the novel environment could have decreased LC axon activity and reduced the magnitude of the novelty signal we detected during running. The novelty signal we report here may therefore be an under estimate of its magnitude under matched behavioral settings.”

However, it is important to note that although VTA axons, on average, showed activity modulated by velocity in a familiar rewarded environment, this relationship was largely due to the activity of two VTA axons that were strongly modulated by velocity, indicating heterogeneity within the VTA axon population in dCA1. We have highlighted this point in the discussion. We also discuss that:

“It is possible that some VTA DA inputs to dCA1 respond to novel environments, and the small number of axons recorded here are not representative of the whole population.”

(4) Sensory properties of the water reward, which the mice may be able to detect, could account for reward-related activity of VTA axons (instead of an expectation of reward). Do the authors have evidence that this is not the case? Occasional probe trials, intermixed with rewarded trials, could be used to test for this possibility.

Mice receive their water reward through a water spout that is immobile and positioned directly in front of their mouth. Water delivery is triggered by a solenoid when the mice reach the end of the virtual track. Therefore, because the water spout is immobile and the water reward is not delivered until they reach the end of the track, there is nothing for the mice to detect during their run. We have added clarifications about the water spout to the Methods and Results sections, along with appropriate discussion points.

Additionally, we note that the ramping activity of VTA axons is still present on the initial laps with no reward (Krishnan et al., 2022), indicating that this activity is not directly related to the presence or absence of water but is instead associated with the animal's reward expectation.

We thank the reviewer for raising this point and hope that these clarifications address their concern.

Reviewer #3 (Public Review):

Summary:

Heer and Sheffield provide a well-written manuscript that clearly articulates the theoretical motivation to investigate specific catecholaminergic projections to dorsal CA1 of the hippocampus during a reward-based behavior. Using 2-photon calcium imaging in two groups of cre transgenic mice, the authors examine the activity of VTA-CA1 dopamine and LC-CA1 noradrenergic axons during reward seeking in a linear track virtual reality (VR) task. The authors provide a descriptive account of VTA and LC activities during walking, approach to reward, and environment change. Their results demonstrate LC-CA1 axons are activated by walking onset, modulated by walking velocity, and heighten their activity during environment change. In contrast, VTA-CA1 axons were most activated during the approach to reward locations. Together the authors provide a functional dissociation between these catecholamine projections to CA1. A major strength of their approach is the methodological rigor of 2-photon recording, data processing, and analysis approaches. These important systems neuroscience studies provide solid evidence that will contribute to the broader field of learning and memory. The conclusions of this manuscript are mostly well supported by the data, but some additional analysis and/or experiments may be required to fully support the author's conclusions.

Weaknesses:

(1) During teleportation between familiar to novel environments the authors report a decrease in the freezing ratio when combining the mice in the two experimental groups (Figure 3aiii). A major conclusion from the manuscript is the difference in VTA and LC activity following environment change, given VTA and LC activity were recorded in separate groups of mice, did the authors observe a similar significant reduction in freezing ratio when analyzing the behavior in LC and VTA groups separately?

In response to the comment regarding the freezing ratios during teleportation between familiar and novel environments, we have analyzed the freezing ratios and lap velocities of DAT-Cre and NET-Cre mice separately (Fig. 3Aiii). Our analysis shows that the mean lap velocities of both groups overlap in the familiar environment and significantly decrease on

the first lap of the novel environment (Fig. 3iii, top). For subsequent laps, the velocities in both groups are not statistically significantly different from the familiar environment lap velocities.

Freezing ratios also show a statistically significant decrease on the first lap of the novel environment compared to the familiar environment in both groups (Fig. 3iii, bottom). In the NETCRE mice, the freezing ratios remain statistically lower in subsequent laps, while in the DATCRE mice, the following laps show a similar trend but without statistical significance. This lack of statistical significance in the DAT-CRE mice is likely due to their already lower freezing ratios in the familiar environment. Overall, the data demonstrate similar behavioral responses in the two groups of mice during the switch from the familiar to the novel environment.

(2) The authors satisfactorily apply control analyses to account for the unequal axon numbers recorded in the LC and VTA groups (e.g. Figure 1). However, given the heterogeneity of responses observed in Figures 3c, 4b and the relatively low number of VTA axons recorded (compared to LC), there are some possible limitations to the author's conclusions. A conclusion that LC-CA1 axons, as a general principle, heighten their activity during novel environment presentation, would require this activity profile to be observed in some of the axons recorded in most all LC-CA1 mice.

We agree with the reviewer's point. To address this issue, when downsampling LC axons to compare to VTA axons, we matched the sampling statistics of the VTA axons/mice by only selecting one LC axon from each mouse to match the VTA dataset.

Additionally, we have now included the number of recording sessions and the number of mice in which we observed each type of activity. This information has been added to further clarify and support our conclusions.

Additionally, if the general conclusion is that VTA-CA1 axons ramp activity during the approach to reward, it would be expected that this activity profile was recorded in the axons of most all VTA-CA1 mice. Can the authors include an analysis to demonstrate that each LC-CA1 mouse contained axons that were activated during novel environments and that each VTA-CA1 mouse contained axons that ramped during the approach to reward?

As above, we have now added the number of mice that had each activity type we report in the paper here.

(3) A primary claim is that LC axons projecting to CA1 become activated during novel VR environment presentation. However, the experimental design did not control for the presentation of a familiar environment. As I understand, the presentation order of environments was always familiar, then novel. For this reason, it is unknown whether LC axons are responding to novel environments or environmental change. Did the authors re-present the familiar environment after the novel environment while recording LC-CA1 activity?

While we did not vary the presentation order of familiar and novel environments, we recorded the activity of LC axons in some mice when exposed to a dark environment (no VR cues) prior to exposure to the familiar environment. Our analysis of this data demonstrates that LC axons are also active following abrupt exposure to the familiar environment.

We have added a new figure showing this response (Supplementary Figure 5A) and expanded on our original discussion point that LC axon activity generally correlates with arousal, as this result also supports that interpretation.

We thank the reviewer for highlighting this important consideration. It certainly helps with the interpretation regarding what LC axons generally encode.

>Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

In addition to what has been described in the public review, I have the following recommendations:

The sample size of DA axon recordings should be increased with the use of a single GCaMP for valid conclusions to be made about the lack of novelty-induced activity in these axons.

We have increased the n of VTA GCaMP6s axons in the familiar environment by including two axons that were recorded in the familiar rewarded condition. We have also conducted an analysis comparing GCaMPs versus GCaMP7b, which is discussed in detail above.

Regarding the concerns about valid conclusions of novelty-induced activity in VTA axons, we have added a comment in the discussion to tone down our conclusions regarding the lack of a novelty signal in the VTA axons. This valid concern is discussed in detail above.

The title is currently very generic, and non-informative. I recommend the use of more specific language in describing the type of behavior under investigation. It is not clear to the reviewer why 'learning' is included here.

Original title: “Distinct catecholaminergic pathways projecting to hippocampal CA1 transmit contrasting signals during behavior and learning”

To make it more specific to the experiments conducted here, we have changed the title to this:

New title: “Distinct catecholaminergic pathways projecting to hippocampal CA1 transmit contrasting signals during navigation in familiar and novel environments”

Error noted in Figure 4C legend - remove reference to VTA ROIs.

The reference to VTA ROIs has been removed from the figure legend

Reviewer #2 (Recommendations For The Authors):

(1) The concluding sentence of the Abstract could be more specific: which distinct types of information are reflected/'signaled'/'encoded' by LC and VTA inputs to dorsal CA1?

The abstract has been adjusted accordingly. The new sentence is more specific: “These inputs encode unique information, with reward information in VTA inputs and novelty and kinematic information in LC inputs, likely contributing to differential modulation of hippocampal activity during behavior and learning.”

(2) Line 46/47: The study by Mamad et al. (2017) did not quite show that VTA dopamine input to dorsal CA1 'drives place preference'. To my understanding, the study showed that suppression of VTA dopamine signaling in a specific place caused avoidance of this place and that VTA dopamine signaling modulated hippocampal place-related firing. So, please consider rephrasing.

Corrected, thanks for pointing this out.

(3) Legend to Figure 3AIII: 'Each lap was compared to the first lap in F . . .' Could you clarify if 'F' refers to the 'familiar environment'?

Figure legend has been changed accordingly

(4) Line 176: '36 LC neurons' - should this not be '36 imaged axon terminals in dorsal CA1' or something along these lines?

This reference has been changed to "LC axon ROIs"

(5) Line 353: Why was water restriction started before the hippocampal window implant, if behavioral training to run for water reward only started after the implant? Please clarify.

A sentence was added to the methods to explain that this was done to reduce bleeding and swelling during the hippocampal window implantation.

(6) Line 377: ' . . . which took 10-14 days (although some mice never reached this threshold).' How many mice did not reach the criterion within 14 days? I think it is not accurate to say the mice 'never' reached the threshold, as they were only tested for a limited period of time.

We have added details of how many mice were excluded from each group and the reason why they were excluded.

(7) Exclusion criteria for imaging data: The authors state (from line 402): 'Imaging sessions with large amounts of drift or bleaching were excluded from analysis (8 sessions for NET mice, 6 sessions for LC Mice).' What exactly were the quantitative exclusion criteria? Were these defined before the onset of the study or throughout the study?

Imaging sessions were first qualitatively assessed by looking for disappearance or movement of structures in the Z-plane throughout the imaging FOV. Additionally, following motion correction in suite2p, we used the registration metrics, which plots the first Principle Component of the motion corrected images, to assess for drift, bleaching, or heat bubbles. If this variable increased or decreased greatly throughout a session, to the point where any apparent activity was not visible in the first PC, the dataset was excluded. We have added these exclusion criteria to the methods section.

Reviewer #3 (Recommendations For The Authors):

Please provide a justification or rationale for having two different criteria for immobility ($< 5\text{cm/sec}$) and freezing ($< 0.2\text{ cm/sec}$). If VTA and LC axon activities are different between these two velocities, please provide some commentary on this difference.

This is a typo leftover from before we converted velocity from rotational units to cm/s.

<https://doi.org/10.7554/eLife.95213.2.sa0>